# PARTICULAR CONDITIONS OF CONTRACT

These Particular Conditions are to be read in conjunction with other documents issued along with the tender. In case of any discrepancy between Design drawings, General conditions or Bill of quantity, more stringent of the same shall be applicable.

The contractor shall refer to the following annexure while bidding and will read them in conjunction with specifications as well as bill of quantity.

| | | |
|---|---|---|
| Annexure - I | : | Design Criterion |
| Annexure -II | : | List of approved makes |
| Annexure -III | : | Codes and Standards |
| Annexure -IV | : | Technical Specifications |
| Annexure -V | : | Technical Data Sheets |

## 1. WORK DESCRIPTION

The work shall be strictly carried out as per the scope listed in this document and in accordance with the specifications. The equipment & material supplied at site will also be selected out of the list of approved makes. Bill of quantity provided with the document is for contractor guidance. It is expected that after award of work, contractor shall prepare shop drawings for approval by the Consultant & Client representative and also submit Technical documentation duly identifying shortlisted make of material/equipment along with its data sheets. Actual ordering shall be based on approved shop drawings & documents.

The work at site shall comply with the approved shop drawings and will meet the satisfaction of Client representative. The contractor shall be required to demonstrate satisfactory operation of entire system (including client supplied equipment installed by contractor) and furnish the required labor, material & tools to install & commission the system.

The broad scope of work for proposed HVAC system covered under this contract shall include supply, installation, testing & commissioning of the following:

- VRF High Side and Low Side System including OEM Indoors
- Copper Piping system and Condensate Drainage
- Mechanical ventilation systems.
- Air distribution system and insulations
- Associated electrical works.
- Testing Adjusting & Balancing of the entire HVAC and ventilation installation.

Besides above, contractor shall also be required to undertake following:

- Obtain approval from client team prior & post installation for operation of system.
- Minor civil works which include making openings in walls and making good of the same.
- Commissioning of the system including test reports to demonstrate satisfactory working prior to handing over.
- Provide as-built drawings and handing over document comprising of list of recommended spares, catalogues and service schedule for each equipment/material.
- Training of Client’s staff.

## 2. SITE MANAGEMENT

The Contractor shall be required to provide following staffing for the project:

a. Design Engineer who will work with consultant for getting shop drawings, technical submittal and variation in quantity statement approved.

b. Procurement team.

c. Full time dedicated Engineer (minimum 10-year experience) & one supervisor posted at site.

d. The contractor shall submit organization chart and CV prior to starting work at site.

The Contractor shall have required stores, tools & plant, security and facility to transport materials to place of installation for speedy execution of work.

**3. REGULATIONS & PERMITS**

Prior to starting work at site, the contractor shall obtain required permits/ licenses required for satisfactory execution and operation of the installation. All receipted amounts shall be reimbursed by Client on production of proof of payment by the contractor.

The executed work shall strictly confirm to applicable laws, regulations and Indian Standards which become applicable. In case the specifications and drawings contained in this document call for higher standard than those required by prevailing regulations, then these specifications & drawings shall become applicable. However, in case of any conflict or violation between the document/drawings and prevailing laws, then the applicable laws & regulations shall be governing & binding.

**4. SHOP DRAWINGS**

A set of design drawings listed in this document are available at Consultant office and may be issued with the tender document. These design drawings are for reference of the contractor and indicate proposed arrangement and the extent of work covered in the contract. The data given in the drawings and specifications is as exact as could be procured, but its accuracy is not guaranteed. The contractor cannot execute work or scale these drawings for reference.

Following shall be the procedure followed by contractor while preparation of shop drawings:

- The contractor shall refer the design drawings for understanding the scope and proposed routes to be followed during execution.

- Collate latest architectural backgrounds from the Client representative / Architect/ Consultant.

- Examine all related services drawings but not limited to structural, plumbing, electrical, HVAC, Interior, landscape and others including as-built works before starting the work. Any discrepancy must be report to the Client's site representative in writing and obtain approval for go-ahead.

- Within one week of award of work, the Contractor shall prepare a list of shop drawing along with submission schedule for approval of Client representative/Consultant. The list of drawings must include layouts for all floors drawings showing exact location of supports, flanges, bends, tee connections, reducers, detailed piping drawings showing exact location and type of supports, valves, fittings etc; electrical panels inside/outside views, power and control wiring schematics, cable trays, supports and terminations.
  Maximum headroom shall be maintained at all points and in case the same is inadequate, then written approval from Client representative must be obtained prior to execution at site.

These shop drawings shall depict information required to complete the Project as per specifications and as required by the Consultant/Client representative. These Drawings shall contain details of construction, size, arrangement, operating clearances, performance characteristics and capacity of all items of equipment, also the details of all related items of work by other contractors. Each shop drawing shall contain tabulation of all measurable items of equipment/materials/works and progressive cumulative totals from other related drawings to arrive at a variation-in-quantity statement at the completion of all shop drawings.

Where the work under this contract is proposed to be installed in close proximity or is interfering with other trades, then based on client representative/consultant directions, the contractor shall prepare all services coordinated working drawings and sections at a suitable scale (not less than 1:50), clearly showing proposed installed in relation to the work of other trades.

- The contractor shall thereafter furnish six sets of detailed shop drawings to Client representative/Consultant for obtaining comments/approval. The Contractor will make unlimited number of re-submissions of shop drawings unless Client representative/Consultant/Architect approval is obtained.

- The Contractor will thereafter submit six sets of final shop drawings to the Client representative for their exclusive use and all other agencies.

- No material or equipment may be delivered or installed at the job site until the contractor has in his possession, the approved shop drawing for the particular material/equipment/installation.

- In case installation is carried out without following above process or obtaining a waiver to follow the procedure from Client representative, the work shall be rejected and contractor shall rectify the same at their own cost.

- Shop drawings shall be submitted for approval a minimum of four weeks in advance of planned delivery and installation of any material to allow Client representative/Consultant ample time for scrutiny. No claims for extension of time shall be entertained because of any delay in the work due to his failure to produce shop drawings at the right time, in accordance with the approved program.

Approval of shop drawings shall not be considered as a guarantee of measurements or of building dimensions. Where drawings are approved, said approval does not mean that the drawings supersede the contract requirements, nor does it in any way relieve the contractor of the responsibility or requirement to furnish material and perform work as required by the contract.

## 5. TECHNICAL DOCUMENTATION

The contractor prior to supplying material at site, will submit the following documentation to Consultant/Client representative for approval:

- Manufacturer's drawings, catalogues, pamphlets and other documents in triplicate. Each item shall be properly labeled, indicating the specific services for which material or equipment is to be used, giving reference to the governing section and clause number and clearly identifying in ink the items and the operating characteristics. Data of general nature shall not be accepted.

- Samples of all materials shall be submitted to the Client's site representative prior to procurement. These will be submitted in two sets for approval and retention by Client's representative and shall be kept in their site office for reference and verification till the completion of the Project. Wherever directed, a mockup or sample installation shall be carried out for approval before proceeding for further installation.

- Where the contractor proposes to use an alternate make or model of equipment other than that specified, all new drawings and detailing required thereafter shall be prepared by the contractor at his own expense including any re-design required for other discipline/trade. Any delay on such account shall also be at the cost of and consequence of the Contractor.

**Contractor to refer Annexure –II for list of approved makes & materials for this project.**

## 6. VARIATION IN QUANTITY STATEMENT

After approval of major & relevant shop drawings, the contractor shall submit four copies of a comprehensive variation in quantity statement. This statement must be submitted prior to completing ordering of equipment and should identify imported/local materials in this contract as well as proposed spares/tools. The Consultant shall provide recommendation to Client representative for acceptance of anticipated variation in contract amounts and also advise Client to initiate action for procurement of spare parts and tools at the completion of project.

## 7. QUALITY ASSURANCE

The contractor to ensure that all materials and equipment supplied shall be new and of best available quality conforming to the relevant Indian Standard Specifications and to these specifications. Makes shall be strictly in conformity with list of approved manufacturers as per Annexure -II. Owner's reserve the right to reject any item which in their assessment is second hand
Any deviations from above shall be clearly highlighted prior to supply and shall be brought to the notice of the Client representative/Consultant for further instructions in the matter.

Prior to starting execution work at site, the Contractor shall verify the sufficiency of the size of the shaft openings, clearances and ceiling spaces for proper installation. Failure to communicate insufficiency of any of the above, shall constitute Contractor acceptance of the same**.** The Contractor shall locate all equipment in fully accessible locations which can be easily serviced, operated or maintained. The exact location and size of access panels, required for each concealed, valve or other devices requiring attendance shall be finalized and communicated in sufficient time. Failing this, the Contractor shall make all the necessary repairs and changes at own expense. Access panel shall be marked.

## 8. WORKS NOT COVERED UNDER THIS CONTRACT

Following works are excluded from the scope under this contract. These shall be executed by respective contractor in accordance with approved shop drawings where these details must be highlighted. However, contractor shall be responsible for providing details and thereafter supervision to ensure satisfactory & timely execution of these associated items as they have a bearing on this contract.

**Civil Works**

- PCC foundation blocks for all air handling units, fans or Outdoor condensing unit
- Water proofing of shafts & slab cutouts required for services.
- Supply and fixing of G.I./wooden frame for mounting of grilles in masonry walls.
- Supply and fixing of GSS frame for mounting of grilles / diffusers in false ceiling / boxing.

**Electrical Works**

- Providing power supply and earthing at the incoming MCCB/MPCB in each air handling unit control panel, fans and VRF outdoor unit.
- Providing 15 amps power outlet within 2-meter reach of each single phase inline fan/ propeller fan, IDU's & ventilation unit at locations called for on air conditioning contractor's shop drawings.

**Plumbing Works**

- Drain for AHU's and IDU's (provision for drain shall be made available by the plumbing contractor and drain pipe till the drain termination shall be under the scope of HVAC contractor.

## 9. TESTING, ADJUSTING AND BALANCING

Air and water balancing shall be carried out by the contractor through a specialist team (different than erection team) as per Specifications and ASHRAE Guide lines. Performance test shall consist of three days of 10 hour each operation of system for each season. The results for each season shall be submitted to Client representative/Consultant. The submittal shall include operational parameters marked on performance curves for each equipment along with test certificates and safety/control settings.

The installation shall be tested again after removal of defects and shall be commissioned only after approval by the Client's site representative. All tests shall be carried out in the presence of the representatives of the Construction Manager / Architect /Consultant and Client's site representative. After commissioning, the results shall be submitted for scrutiny in quadruplicate.

All equipment shall operate under all conditions of load without any sound or vibration which is objectionable in the opinion of the Client's site representative. In case of rotating machinery sound or vibration noticeable outside the room in which it is installed, or annoyingly noticeable inside its own room, shall be considered objectionable. Such conditions shall be corrected by the Contractor at his own expense. The contractor shall guarantee that the equipment installed shall maintain the specified Noise Control levels.

## 10. COMPLETION CERTIFICATE

On completion of the installation, a certificate shall be furnished by the contractor, counter signed by the licensed supervisor, under whose direct supervision the installation was carried out. This certificate shall be in the prescribed form as required by the local authority.

The contractor shall be responsible for getting the entire installation duly approved by the local authorities concerned, and shall bear expenses if any, in connection with the same.

## 11. AS-BUILT DRAWINGS

Contractor shall submit following as-built drawings as and when work is completed:

- Six set of hard copies of all as-built drawings duly corrected and incorporating any modifications during execution.
- Two set of pen drive containing the drawings.

The drawings shall provide AHU layouts, piping layouts, location of all concealed accessories/piping, wiring diagram, control diagram, Single line diagram, control schematic with detailed bill of materials, showing makes, types & description of all components & accessories and sequencing of automatic controls and other services.

## 12. MAINTENANCE MANUAL

Upon completion and commissioning of works, the contractor shall submit a draft copy of comprehensive operating instructions, maintenance schedule and log sheets for all systems and equipment included in this contract. This shall be supplementary to manufacturer's operating and maintenance manuals. Upon approval of the draft, the contractor shall submit four (4) complete bound sets of typewritten operating instructions and maintenance manuals; one each for retention by Consultant and Client's site representative and two for Clients Operating Personnel. These manuals shall also include basis of design, detailed technical data for each piece of equipment as installed, spare parts manual and recommended spares for 4 year period of maintenance of each equipment.

The manuals shall include:

i. Description of the work carried out / installed.

ii. Operating instructions.

iii. Maintenance instructions including procedures for preventive maintenance.

iv. Manufacturers catalogues.

v. Spare parts list.

vi. Trouble shooting charts.

vii. Drawings

viii. Type and routine test certificates of major items.

Details of all the bought-out item should be part of this maintenance manual.

**13. ON SITE TRAINING**

Upon completion of all work and all tests, the Contractor shall furnish necessary operators, labor and helpers for operating the entire installation for such periods so as to enable the Client's staff to get acquainted with the operation of the system. During this period, the contractor shall train the Client's personnel in the operation, adjustment and maintenance of all equipment installed.

**14. DEFECTS LIABILITY PERIOD**

**Complaints**

The Contractor shall receive calls for any and all problems experienced in the operation of the system under this contract, attend to these within 10 hours of receiving the complaints and shall take steps to immediately correct any deficiencies that may exist.

**Repairs**

All equipment that requires repairing shall be immediately serviced and repaired. Since the period of Mechanical Maintenance runs concurrently with the defects liability period, all replacement parts and labour shall be supplied promptly free-of-charge to the Client.

**15. UPTIME GUARANTEE**

The contractor shall guarantee for the installed system an uptime of 98%. In case of shortfall in any month during the defects liability period, the Defects Liability period shall get extended by a month for every month having shortfall and no reimbursement shall be made for the extended period.

**16. OPERATION & MAINTENACE CONTRACT**

Contractor may be required to carry out the operation of the installation during and after the defect's liability period. Further, it may also be required to carry out all-inclusive maintenance of the entire system for a period of four years beyond the defects liability period.

**Operation Contract:**

It will involve round the clock operation for 24 hours a day wherein work will include but not limited to operation of installation, maintaining log books, complain register and summary of operation.

The terms of payment shall be monthly at the end of each month on pro-rata basis.

**All Inclusive Maintenance Contract:**

The work will involve routine preventive maintenance with monthly status report. Entire installation shall be painted every two years. 98% uptime of all systems is expected under this contract wherein up time shall be assessed every month and in case of shortfall during any month the contract shall be extended by a month. No reimbursement shall be payable for the extended period.
Adequate number of persons to the satisfaction of the Client representative shall be provided including relievers wherein statutory compliances such as of EPF, ESIC and other applicable labour legislations shall be to contractor account. No overtime shall be payable. Routine shut downs shall be permitted with prior permission of the Owner.

Payment shall be Quarterly at the beginning of each quarter on pro-rata basis.

# ANNEXURE-I

## DESIGN CRITERION

### 1 DESIGN CRITERION

Following shall be basis for developing the design:

| | | |
|---|---|---|
| Site Location | : | Ahmedabad |
| Geographical Data | : | 23.04’ N, 72.37’ E |
| Altitude | : | 55 m above mean sea level |

### 1.1 Outdoor Design Temperatures

Recommended outdoor design conditions have been defined in ISHRAE Weather Data File 2017 which has been published jointly with Bureau of Energy Efficiency (BEE) for Ahmedabad.

The HVAC design for the entire facility shall be based on 0.4% annual cumulative frequency of occurrence for Summer & Monsoon and 99.6% annual cumulative frequency of occurrence for winter.

| S No. | Season | Outdoor Temperatures | |
|---|---|---|---|
| | | DBT | WBT |
| 1. | Summer | 43.1 Deg.C (109.5 °F) | 23 Deg C (73.4 °F) |
| 2. | Monsoon | 33.8 Deg C (92.84 °F) | 28.6 Deg.C (83.4 °F) |
| 3. | Winter | 10.9 Deg.C (57.9 °F) | - |

### 1.2 Envelope Details

Based on the site review, the performance parameter of envelope is given below:

| S. No. | Space | Construction details | Thermal Performance |
|---|---|---|---|
| 1. | Exposed Masonry Walls | 230 mm thick Brick Wall | 1.45 Watt / Sqm °C (0.25 Btu / Hr. Sft °F) |
| 2. | Exposed Roof (Insulated | Min. 32mm thick Nitrile rubber Thermal under deck insulation shall be carried out for all roofs (above conditioned areas) exposed to sun. | U’ Value = 0.12 BTU / Hr. - Sft |
| 3. | Glazing | Single Glazing Unit | 2.0 Watt / Sq m °C (0.65 Btu / Hr. Sft °F) SHGC: 1.2 |

***Notes-***

- *The roof receives significant solar radiation and plays an important role in heat gain/loss, day lighting and ventilation. Depending on the climatic needs, proper roof treatment is essential. The roof shall comply with either the maximum assembly U factor or the minimum insulation R-value.*
- *The above envelope details are verified by the architectural team and the same has been confirmed by Adani architecture team.*

## 2.0 HVAC SYSTEM

### 2.1 Indoor Design Conditions

In line with the design standards and prevalent design practices, following indoor design condition is proposed.

| S. No. | Description | Temperature °C | Relative Humidity |
|---|---|---|---|
| **A.** | **Block 1** | | |
| 1. | Dormitory / Double Occupancy Room | Non-AC | No Control |
| 2. | Recreational Room | 23± 1 °C | Less Than 60% |
| 3. | Dining Area | Non-AC | No Control |
| 4. | Lift Lobby | Non-AC | No Control |
| 5. | Corridor | | |
| **B.** | **Block 2** | | |
| 1. | Double Occupancy Room | Non-AC | No Control |
| 2. | Corridor | | |
| 3. | Lift Lobby | | |
| **C.** | **Block 3** | | |
| 1. | Dormitory / Double Occupancy Room | Non-AC | No Control |
| 2. | Recreational Room | 23± 1 °C | < 60% |
| 3. | Lift Lobby and Corridor | Non-AC | No Control |
| **D.** | **Block 4** | | |
| 1. | Ammunition Store/ Kote/Issue Area | Non-AC | No Control |
| 2. | Workshop | | |
| 3. | Guard Room | | |
| **E.** | **Block 5 (Kennel)** | **23± 1 °C** | **< 60%** |

### 2.2 Outdoor Air Ventilation rate in breathing zones

| Space | | Fresh Air Ventilation Rate |
|---|---|---|
| Single/Double Rooms | : | Outdoor air shall leak into the occupied spaces through undercuts & door opening |
| Dormitory | : | Outdoor air shall leak into the occupied spaces through undercuts & door |

### 2.3 Ventilation Rates

| S. No | Space | Air Changes Per Hours (ACPH) |
|---|---|---|
| 1. | Public Toilet | 10 ACPH exhaust with equivalent draw of air from Door Undercut/Door transfer Grille. |
| 2. | Room Washrooms | 6 ACPH exhaust with equivalent draw of air from Door Undercut/Door transfer Grille. |
| 4. | Car parking (Basement) | Min. 6 ACPH exhaust with equivalent draw of make-up Air (Normal) 12 ACPH exhaust with equivalent draw of make-up Air (Fire) |
| 5. | Kitchen | Min. 40 ACPH with slightly lesser draw of makeup air to maintain Negative Pressure Inside the kitchen. |
| 6. | Pantry | 6 ACPH exhaust with equivalent draw of air from adjacent areas |
| 7. | E&M Plantroom | 15 ACPH exhaust with equivalent draw of air from adjacent areas |
| 8. | Kennels | 10 ACPH exhaust with equivalent draw of air from adjacent areas. |
| 9. | Laundry | 15 ACPH exhaust with equivalent draw of air from adjacent areas |

### 2.4 Pressurization of Lift Well, Staircase and Lift Lobbies

| S. No. | Area | Pressurization Status | Remarks |
|---|---|---|---|
| 1 | Lift Lobbies | N/A | Lift lobbies are opening in the corridor |
| 2 | Staircases | Naturally Ventilated | All Staircases are toward the façade |
| 3 | Lift Wells | Not Pressurized | Refer NBC Vol 1, Part 4 * |

As the doors of lift lobbies are open it doesn't create any flue effect attached snap for NBC. 2016.

pressurization is required for lift commuting from ground to basement

NOTES

**1** The natural ventilation requirement of the staircase shall be, achieved through opening at each landing, of an area 0.5 $m^2$ in the external wall. A cross ventilated staircase shall have 2 such openings in opposite/adjacent walls or the same shall be cross-ventilated through the corridor.

**2** Enclosed staircase leading to more than one basement shall be pressurized.

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[1)] Lift lobby with fire doors (120 min) at all levels with pressurization of 25-30 Pa is required. However, if lift lobby cannot be provided at any of the levels in air conditioned buildings or in internal spaces where funnel/flue effect may be created, lift hoistway shall be pressurized at 50 Pa. For building greater than 30 m, multiple point injection air inlets to maintain desired pressurization level shall be provided. If the lift lobby, lift and staircase are part of firefighting shaft, lift lobby necessarily has to be pressurized in such case, unless naturally ventilated.

## 2.0 Heat Load Calculation

| S.NO. | Description | Area (Sq Ft) | Occupancy | Outdoor Air | Lighting load | Equipment load | Dehumidified CFM | Total TR (Summer) | Total TR (Monsoon) | Equipment Selection |
|---|---|---|---|---|---|---|---|---|---|---|
| **A.** | **Block -1** | | | | | | | | | |
| 1 | Recreational | 829 | 25 | 175 | 829 | 829 | 1921 | 4.3 | 4.4 | 4 No. 1.5 TR DX Cassette Unit |
| | **Sub Total** | **829** | **25** | **175** | **829** | **829** | **1921** | **4** | **4** | |
| **B.** | **Block -3 Ladies Barrack** | | | | | | | | | |
| 1 | Recreational | 366 | 12 | 82 | 366 | 366 | 912 | 2.0 | 2.1 | 2 No. 1.5 TR DX Cassette Unit |
| | **Sub Total** | **366** | **12** | **82** | **366** | **366** | **912** | **2** | **2** | |
| **C.** | **Block -3 Kennels** | | | | | | | | | |
| 1 | Isolation kennel 1 | 48 | 1 | 8 | 48 | 48 | 139 | 0.28 | 0.25 | 1 No. 1.0 TR Hi Wall Unit |
| 2 | Isolation kennel 2 | 48 | 1 | 8 | 48 | 48 | 95 | 0.20 | 0.22 | 1 No. 1.0 TR Hi Wall Unit |
| 3 | Kennel 1 | 48 | 1 | 8 | 48 | 48 | 95 | 0.20 | 0.22 | 1 No. 1.0 TR Hi Wall Unit |
| 4 | Kennel 2 | 48 | 1 | 8 | 48 | 48 | 95 | 0.20 | 0.20 | 1 No. 1.0 TR Hi Wall Unit |
| 5 | Kennel 3 | 48 | 1 | 8 | 48 | 48 | 95 | 0.20 | 0.22 | 1 No. 1.0 TR Hi Wall Unit |
| 6 | Kennel 4 | 48 | 1 | 8 | 48 | 48 | 95 | 0.20 | 0.22 | 1 No. 1.0 TR Hi Wall Unit |
| 7 | Kennel-5 | 48 | 1 | 8 | 48 | 48 | 158 | 0.32 | 0.28 | 1 No. 1.0 TR Hi Wall Unit |
| 8 | Isolation kennel 3 | 48 | 1 | 8 | 48 | 48 | 127 | 0.26 | 0.24 | 1 No. 1.0 TR Hi Wall Unit |
| 9 | Isolation kennel 4 | 48 | 1 | 8 | 48 | 48 | 95 | 0.20 | 0.22 | 1 No. 1.0 TR Hi Wall Unit |
| 10 | Isolation kennel 5 | 48 | 1 | 8 | 48 | 48 | 95 | 0.20 | 0.22 | 1 No. 1.0 TR Hi Wall Unit |
| 11 | Kennel 6 | 48 | 1 | 8 | 48 | 48 | 95 | 0.20 | 0.22 | 1 No. 1.0 TR Hi Wall Unit |
| 12 | Kennel 7 | 48 | 1 | 8 | 48 | 48 | 95 | 0.20 | 0.22 | 1 No. 1.0 TR Hi Wall Unit |
| 13 | Kennel 8 | 48 | 1 | 8 | 48 | 48 | 95 | 0.20 | 0.22 | 1 No. 1.0 TR Hi Wall Unit |
| 14 | Kennel 9 | 48 | 1 | 8 | 48 | 48 | 95 | 0.20 | 0.22 | 1 No. 1.0 TR Hi Wall Unit |
| | **Sub Total** | **678** | **14** | **111** | **678** | **678** | **1466** | **3** | **3** | **16 HP VRF UNIT** |
| | **Grand Total** | **1873** | **51** | **367** | **1873** | **1873** | **4299** | **9** | **10** | |

### 2.0.1 Smoke Extraction & Utility Ventilation

All independent / sharing rooms in Block 1, Block 2 and Block 3 Shall be provided with 1 No. 230mm dia Propeller fan in addition to the following fans-

| S. No. | Description | Area Sqm | Area Sft. | Height Ft. | Volume Cuft. | Ventilation Rate ACPH | Application | Air quantity Calculated CFM | Equipment Selection |
|---|---|---|---|---|---|---|---|---|---|
| **A.** | **Block -1** | | | | | | | | |
| | **Basement** | | | | | | | | |
| 1 | Car Parking Zone -1 | 570 | 6123 | 11.0 | 67350 | 12 | Make-up air | 13470 | 2.No 7000 CFM Axial Flow Fan |
| | | 570 | 6125 | 11.0 | 67377 | 12 | Exhaust | 13475 | 2.No 7000 CFM Axial Flow Fan (Fire Rated) |
| 2 | Car Parking Zone -2 | 570 | 6123 | 11.0 | 67350 | 12 | Make-up air | 13470 | 1.No 7000 CFM Axial Flow Fan(50% Fresh air Through Ramp) |
| | | 570 | 6125 | 11.0 | 67377 | 12 | Exhaust | 13475 | 2.No 7000 CFM Axial Flow Fan (Fire Rated) |
| 2 | Laundry | 115 | 1235 | 11.0 | 13588 | 15 | Make-up air | 3397 | 1.No 3000 CFM Fan section Unit |
| | | 115 | 1236 | 11.0 | 13594 | 15 | Exhaust | 3398 | 1.No 3500 CFM Fan section Unit |
| | **Ground Floor** | | | | | | | | |
| 1 | Laundry | 40 | 430 | 10.0 | 4297 | 15 | Make-up air | 1074 | Make of through Doors |
| | | 40 | 430 | 10.0 | 4298 | 15 | Exhaust | 1075 | 1 No. 350 Dia Propeller Fan |
| 2 | Kitchen | 48 | 516 | 10.0 | 5156 | 40 | Make-up air | 3437 | 1 No.3200CFM Cabinet Fan |
| | | 48 | 516 | 10.0 | 5158 | 40 | Exhaust | 3439 | 1 No.3500 CFM Cabinet Fan |
| 3 | Toilet -1 | 24 | 258 | 10.0 | 2578 | 10 | Exhaust | 430 | 1 No. 500 CFM Inline Fan |
| | Toilet -2 | 20 | 215 | 10.0 | 2148 | 10 | Exhaust | 358 | 1 No. 500 CFM Inline Fan |
| **A.** | **Ladies Barrack** | | | | | | | | |
| 1 | Ground Floor Toilet | 22 | 234 | 10.0 | 2342 | 10 | Exhaust | 390 | 1 No. 400 cfm Inline Fan |
| **A.** | **Kennels Block-5** | | | | | | | | |
| 1 | Kennels -1 | 100 | 1074 | 10.0 | 10742 | 10 | Exhaust | 1790 | 1 No.1800 CFM Cabinet Fan |
| 2 | Kennels -2 | 100 | 1074 | 10.0 | 10742 | 10 | Exhaust | 1790 | 1 No.1800 CFM Cabinet Fan |

## 2.1 System Description

### 2.1.1 Block -1 2 & 3

Considering the nature of the building, it is proposed to consider electrical load provision for Block-1, Block 2 & Block 3 which shall be air conditioned through conventional inverter type split air conditioners of suitable size (as and when required). The system will comprise of multiple evaporators (indoors) and multiple outdoor condensing unit planned on Facade or dedicated space for outdoor.

Whereas areas such as recreational zones, dining area has been considered completely air conditioned through the same system, wherein indoor type shall be as per the interior intent.

### 2.1.2 Kennel room Air conditioning.

Air-conditioning through VRF shall be provided for the development. System comprising of one or more air cooled outdoor unit and multiple indoor units. The outdoor unit shall house multiple compressors and be located at terrace or any other dedicated service area. The indoor unit may be hi-wall type. Indoor units shall be connected to outdoor unit through a single copper refrigerant pipe system. One or more compressors in each outdoor unit shall be provided with variable frequency drive whereby refrigerant flow through copper pipe shall be varied based on the AC load being encountered. The outdoor unit shall have built-in energy efficiency features like capacity control, oil return operation controls, intelligent defrost control and compressor control etc.

The indoor units shall be similar in operation and appearance as conventional split units and shall provide facility for independent on-off control, temperature setting etc. The VRF system has following inherent advantages over conventional split unit system

Individual accurate space control

Multiple compressors in outdoor unit in conjunction with inverter drive compressor to modulate refrigerant flow based on requirement.

Minimizing heat transfer losses due to superior refrigerant piping system.

The Variable Refrigerant Flow (VRF) system shall provide additional benefits in terms of low sound levels due to remote compressor location. Temperature setting of each indoor unit shall be controllable through individual cordless micro-processor based remote unit that has self-diagnostic facility to automatically identify fault for more reliable operation.

The noise criterion for the proposed VRF shall be not exceeding 45 dB(A) for indoor units and 65 dB(A) for outdoor units at 3 m distance.

## 2.2 Toilet Exhaust

Individual Toilets are ventilated the ventilator and public area toilet exhaust arrangement shall be provided through Inline Fan. For Individuals toilets only electrical points shall be provided for the fan.

### 2.3 Basement Ventilation Block -1

Car park areas shall be designed to permit 12 ACPH of smoke extraction in case of emergency. The basement shall be divided into two zones with each zone size not exceeding 3000 sqm.

All Supply air and exhaust air fans shall be installed in fire resisting room of 120 min as per guidelines of latest NBC 2016.

Ducted Ventilated system is envisaged for the project wherein vertical risers shall be planned independently for makeup air and exhaust air connected to dedicated fan through a duct. Axial fans for basement ventilation shall be provided with VFD which shall be interlinked with CO levels inside the car parking area, these fans shall ramp up to suitable air volume. The system shall maintain at least 6 ACPH at all times as per the requirement of NBC 2016.

# Annexure-II

## List of Approved Makes

| S. No | Equipment/Material | Approved Manufacturer Name |
|---|---|---|
| 1 | Variable Refrigerant System | Daikin<br>Toshiba<br>Mitsubishi |
| 2 | Inverter Split Units/ Split AC | Daikin<br>Toshiba<br>Mitsubishi<br>Bluestar |
| 3 | Cabinet/ Inline/ Propeller Fan | Kruger<br>System air<br>Nicotra |
| 4 | Refrigerant & Drainpipe tube Insulation | Armacell<br>ALP Aeroflex<br>Aerofoam |
| 5 | Copper Pipe | Mandev<br>Total line<br>RR shramik |
| 6 | U PVC pipe | Finloex<br>Astral<br>Ashirwad |
| 7 | Centrifugal Fan (AMCA certified) | Greenheck<br>Systemair<br>Nicotra |
| 8 | Axial Fans (AMCA Certified) for air and sound performance | Greenheck<br>System air<br>Nicotra<br>Kruger |
| 9 | Propeller Fan | Cary Aire<br>Nicotra<br>Humidin |
| 10 | Inline Fan (Super Slient Mixed flow fan) | Kruger<br>System Air<br>Greenheck |
| 11 | AL/GI Sheets (Lead free) | Bhushan<br>Essar<br>Jindal<br>Lloyd<br>SAIL<br>TATA |
| 12 | Factory Fabricated ducts | GPSpira<br>7star<br>Ductofab |

| S. No | Equipment/Material | Approved Manufacturer Name |
|---|---|---|
| 13 | Equipment/ duct supports | Gripple<br>Hilti<br>Fischer |
| 14 | Grille/Diffuser/Dampers | Conaire<br>Tristar<br>Systemaire<br>Cosmos |
| 15 | Duct Insulation | Armacell<br>ALP<br>Aeroflex |
| 16 | Duct Lining | Armacell<br>ALP<br>Aeroflex |
| 17 | Vibration Isolator | Dunlop<br>Easyflex<br>Flexionics<br>Resistoflex |
| 18 | Anchor Fastener | Fischer<br>Hilti<br>Hira Walraven |
| 19 | Motor | BharatBijlee<br>ABB<br>Siemens |
| 20 | Electrical Panels | Adlec Control System<br>Ambit<br>Neptune |
| 21 | MCB/RCCB/ MCCB/ MPCB | Siemens<br>Schneider Electric<br>L&T<br>ABB |
| 22 | Indicating Lamps LED type and Push Button | Schneider Electric<br>Siemens<br>ESBEE<br>Teknik Vaishno |
| 23 | LT Cables | Polycab<br>KEI<br>RPG Cables<br>Skytone |
| 24 | Termination kits | 3M<br>Raychem |
| 25 | Double Compression Cable Glands with earthing links | Dowells<br>Comet |
| 26 | Bimetallic Cable Lugs | Dowells<br>Comet<br>Cosmos |

| S. No | Equipment/Material | Approved Manufacturer Name |
|---|---|---|
| 27 | PVC Insulated FRLS Wires | RR Kabel<br>KEI<br>Skytone<br>Bonton |
| 28 | Cable Trays & Raceways | Ricco<br>MEM<br>Profab |
| 29 | Metal Conduit & Accessories | BEC<br>AKG |
| 30 | PVC Conduit & Accessories | BEC<br>AKG<br>Polypack<br>Precision |
| 31 | Control Transformer / Potential | Automatic Electric |
| 32 | Digital Meters | Schneider Electric<br>L&T<br>Siemens |
| 33 | Starter | Schneider<br>ABB<br>Siemens |
| 34 | GI/MS Conduit | AKG<br>BEC |

# ANNEXURE-III

## PART LIST OF CODES & STANDARDS

The installation in entirety shall comply with latest codes/standards published by Bureau of Indian Standards (BIS) as well as local regulations from departments like Pollution Control Board, Electrical inspectorate, Fire Authorities, Airport Authority of India (AAI), High rise committee, Indian Electricity rules etc. Some of the standards are mentioned here below for reference:

| | |
|---|---|
| ASHRAE Handbooks | Systems & Equipment 2012.<br>Fundamentals 2013.<br>Refrigeration 2014.<br>Application 2015. |
| | ASHRAE Standard 62.1-2013.<br>ASHRAE Standard 90.1-2013<br>ASHRAE Standard 55-2010<br>ASHRAE Standard 52.1 and 52.2 |
| | Energy Conservation Building code of India -2008 (BEE) |
| IS : 277 - 1992 | Galvanized steel sheet (Plain & Corrugated) wire for fencing. |
| IS : 554 - 1985 (Reaffirmed 1996) | Dimensions for pipe threads where pressure tight joints are required on the threads. |
| IS : 655 - 1963 (Reaffirmed 1991) | Metal air ducts. |
| IS : 659 – 1964 (Reaffirmed 1991) | Air conditioning (Safety Code) |
| IS : 660 – 1963 (Reaffirmed 1991) | Mechanical Refrigeration (Safety Code) |
| IS : 694 - 1990 (Reaffirmed 1994) | PVC insulated (HD) electric cables for working voltage upto and including 1100 volts. |
| IS : 732 - 1989 | Code of practice for electrical wiring. |
| IS : 780 - 1984 | Sluice valves for water works purposes. |
| IS : 822-1970 (Reaffirmed 1991) welds. | Code of procedure for inspection of |
| IS : 1239 (Part - I) - 1990 | Mild steel tube |
| IS : 1239 (Part - II) - 1992 | Mild steel Tubulars and other wrought steel pipe fittings. |
| IS : 1255 - 1983 | Code of Practice for installation and maintenance of Power Cables |

| | |
|---|---|
| IS 1554 - 1988 (Part – I) | PVC insulated ( Heay Duty) electric cables for working voltages pto and including 1100 volts. |
| IS 1897 – 1983 (Reaffirmed 1991) | Copper bus bar / strip for electrical purposes |
| IS 2379 - 1990 | Colour code for the identification of pipelines. |
| IS 2551 - 1982 | Danger notice plate |
| IS 3043 - 1987 | Code of practice for earthing. |
| IS 3103 – 1975 (Reaffirmed 1999) | Code of practice for Industrial Ventilation. |
| IS 3837 – 1976 (Reaffirmed 1990) | Accessories for rigid steel conduit for electrical wiring. |
| IS 4736 – 1986 (Reaffirmed 1998) | Hot-dip zinc coatings on steel tubes. |
| IS 4894 - 1987 | Centrifugal Fan. |
| IS 5133 - 1969 (Part-I) (Reaffirmed 1990) | Boxes for the enclosure of electrical accessories. |
| IS 5216 – 1982 (Part-I) (Reaffirmed 1990) | Guide for safety procedure and practices in electrical work. |
| IS 5312 (Part-I) - 1984 (Reaffirmed 1990) | Swing - check type reflux Non return valves for water works |
| IS 5424 – 1989 (Reaffirmed 1994) | Rubber mats for electrical purposes. |
| IS : 5578 & 11353-1985 | Marking and identification of conductors |
| IS : 6392 - 1971(Reaffirmed 1988) | Steel pipe flanges. |
| IS : 8623 - 1993 | Low voltage switchgear and control gear Assemblies (Requirement for type / partly type tested assemblies) |
| IS : 8623 - 1993 | Bus Bar trunking system |

| | |
|---|---|
| IS : 8828 - 1996 | Circuit Breakers for over current protection For house hold and similar installation. |
| IS : 9537 - 1981 (Part II) | Rigid Steel Conduits for electrical wiring |
| IS : 10810 - 1988 | Methods of test for cables. |
| IS : 13947-1993 (Part-I) | General rules for low voltage switch gears and control gears. |
| IS : 13947-1993 (Part-II) | Circuit BreakersIEC 947 - 2 |
| | IS : 13947 - 1993 (Part-III) Swiches, disconnectors and fuse for low voltage switch gear and control gear. |
| IS : 13947 - 1993 (Part-IV) | Low voltage switch gear and controlgear for contactor and motor starters |
| IS : 13947 – 1993 (Part-V) | Control Circuit Devices. |

# ANNEXURE-IV
# TECHNICAL SPECIFICATIONS

## 1.0 VARIABLE REFRIGERANT VOLUME SYSTEM

### 1.1 Scope

The scope of this section comprises the supply, erection testing and commissioning of Variable Refrigerant Volume System conforming to these specifications and in accordance with the requirements of Drawing and Schedule of Quantities

### 1.2 Type

Units shall be air cooled, variable refrigerant volume air conditioner consisting of one outdoor unit and multiple indoor units. Each indoor units having capability to cool or heat independently for the requirement of the rooms.

It shall be possible to connect minimum 10 indoor units on one refrigerant circuit. The indoor units on any circuit can be of different type and also controlled individually. Following type of indoor units shall be connected to the system:

- Ceiling mounted Ductable type
- Ceiling suspended type
- Wall mounted type

Compressor installed in outdoor unit shall be equipped with inverter controller, and capable of changing the rotating speed to follow variations in cooling and heating load.

Outdoor unit shall be suitable for mix match connection of all type of indoor units.

The refrigerant piping between indoor units and outdoor unit shall be extended up to 150m with maximum 50m level difference **without any oil traps.**

Both indoor units and outdoor unit shall be factory assembled, tested and filled with first charge of refrigerant before delivering at site.

### 1.3 Outdoor Unit

The outdoor unit shall be factory assembled, weather proof casing, constructed from heavy gauge mild steel panels and coated with baked enamel finish. The unit should be completely factory wired , tested with all necessary controls and switch gears:

- All outdoor units shall have 100% Inverter compressors and be able to operate even in case one of compressor is out of order.
- In case of outdoor units with multiple compressors, the operation shall not be disrupted with failure of any compressor.
- It should also be provided with duty cycling for switching starting sequence of multiple outdoor units.
- The noise level shall not be more than 65 dB(A) at normal operation measured horizontally 1m away and 1.5m above ground level.
- The outdoor unit shall be modular in design and should be allowed for side by side installation
- The unit shall be provided with its own microprocessor control panel.

The outdoor unit should be fitted with low noise, aero spiral design fan with large airflow and should be designed to operate compressor linking technology. The unit should also be capable to deliver 55 Pa external static pressure to meet long exhaust duct connection

**1.4 Compressor**

The compressor shall be highly efficient scroll type and capable of inverter control. It shall change the speed in accordance to the variation in cooling or heating load requirement:

- The inverter shall be IGBT type for efficient and quiet operation.
- All outdoor units shall have at least 10 steps of capacity control to meet load fluctuation and indoor unit individual control. All parts of compressor shall be sufficiently lubricated stock. Forced lubrication may also be employed.
- Oil heater shall be provided in the compressor casing.

**1.5 Refrigerant circuit**

The refrigerant circuit shall include liquid & gas shut-off valves and a solenoid valves at condenser end.

**1.6 Safety Services**

All necessary safety devices shall be provided to ensure safe operation of the system.

Following safety devices shall be part of outdoor unit; high pressure switch, fuse, crankcase heater, fusible plug, over load relay, protection for inverter, and short recycling guard timer.

**1.7 Oil Recovery System**

Unit shall be equipped with an oil recovery system to ensure stable operation with long refrigeration piping lengths.

**1.8 Indoor Unit**

This section deals with supply, installation, testing, commissioning of various type of indoor units confirming to general specification and suitable for the duty selected. The type, capacity and size of indoor units shall be as specified in detailed Bill Of Quantities

**1.9 General**

Indoor units shall be either ceiling mounted cassette type, or ceiling mounted ductable type or floor standing type or wall mounted type or other as specified in BOQ. These units shall have electronic control valve to control refrigerant flow rate respond to lead variations of the room.

a. The address of the indoor unit shall be set automatically in case of individual and group control

b. In case of centralized control, it shall be set by liquid crystal remote controller

The fan shall be dual suction, aerodynamically designed turbo, multi blade type, statically & dynamically balanced to ensure low noise and vibration free operation of the system. The fan shall be direct driven type, mounted directly on motor shaft having supported from housing. The cooling coil shall be made out of seamless copper tubes and have continuous aluminum fins. The fins shall be spaced by collars forming an integral part. The tubes shall be staggered in the direction of airflow. The tubes shall be hydraulically/ mechanically expanded for minimum thermal contact resistance with fins. Each coils shall be factory tested at 21kg/sqm air pressure under water.

Unit shall have cleanable type filter fixed to an integrally moulded plastic frame. The filter shall be slide away type and neatly inserted.

Each indoor unit shall have computerized PID control for maintaining design room temperature. Each unit shall be provided with microprocessor thermostat for cooling and heating.

Each unit shall be with wired LCD type remote controller. The remote controller shall memorize the latest malfunction code for easy maintenance. The controller shall have self-diagnostic features for easy and quick maintenance and service. The controller shall be able to change fan speed and angle of swing flat individually as per requirement.

**1.10.1** Ceiling Mounted Ductable Type Indoor Unit

Unit shall be suitable for ceiling mounted type. The unit shall include pre filter, fan section & DX coil section. The housing of unit shall be light weight powder coated galvanized steel. The unit shall have high static fan for Ductable arrangement.

**1.10.2** Ceiling Suspended Indoor Type

Unit shall be suitable for ceiling suspended arrangement below false ceiling. The unit include pre filter, fan section & DX coil section. The housing of unit shall be light weight powder coated galvanized steel.

**1.10.3** High Wall Mounted Indoor Units

The units shall be wall-mounted type. The unit includes pre filter, fan section & DX coil section. The housing of unit shall be light weight powder coated galvanized steel.

Unit shall have an attractive external casing for supply and return air.

**1.10.4** Centralized Type Remote Controller (if Specified In BOQ)

A multifunctional compact centralized touch screen type controller shall be provided with the system.

The Graphic Controller must act as an advanced air-conditioning management system to give complete control of VRV air-conditioning Equipment, It should have ease of use for the user through its touch screen, icon display and color LCD display.

It shall be able to control up to 64 groups of indoor units with the following functions :-

a) Starting/stopping of Air conditoners as a zone or group or individual unit. Temperature settling for each indoor unit or zone.

b) Switching between temperature control modes, switching of fan speed and direction of airflow, enabling/disabling of individual remote controller operation.

c) Monitoring of operation status such as operation mode & temperature setting of individual indoor units, maintenance information, troubleshooting information.

d) Display of air conditioner operation history.

e) Daily management automation through yearly schedule function with possibility of various schedules.

The controller shall have wide screen user friendly color LCD display and can be wired by a non-polar 2 wire transmission cable to a distance of 1 km. away from indoor unit.

**Performance of VRF Condensing Unit**

Following shall be the required COP for the given condensing unit for both cooling and monsoon

| Model | Type | COP - Cooling @ 100% Load | COP - Heating @ 100% Load |
|---|---|---|---|
| 38 HP | Top Discharge | 4.5 | 4.7 |
| 36 HP | Top Discharge | 4.1 | 4.55 |
| 30 HP | Top Discharge | 3.5 | 4.02 |
| 26 HP | Top Discharge | 4.0 | 4.37 |
| 22 HP | Top Discharge | 3.75 | 4.10 |
| 14 HP | Top Discharge | 4.21 | 4.70 |
| 10 HP | Top Discharge | 4.51 | 5.32 |
| 6 HP | Side Discharge | 3.43 | 3.56 |

## 3.0 FANS

The scope of work shall comprise of Supply, installation, testing & commissioning of Fans as per the requirement specified in tender BOQ as well as drawings.

### 3.1 Centrifugal Fans

Centrifugal fan shall be DWDI / SWSI Class I construction arrangement 3 (i.e. bearings on both the sides) for DWDI fans complete with access door, squirrel-cage induction motor, V-belt drive, belt guard and vibration isolators, direction of discharge / rotation, and motor position shall be as per the Approved-for-Construction shop drawings.

Housing shall be constructed of 14 gage sheet steel welded construction. It shall be rigidly reinforced and supported by structural angles. Split casing shall be provided on larger sizes of fans, however neoprene / asbestos packing should be provided throughout split joints to make it air-tight.

18 gauge galvanized wire mesh inlet guards of 5 cm sieves shall be provided on both inlets. Housing shall be provided with standard cleanout door with handles and neoprene gasket. Rotation arrow shall be clearly marked on the housing.

Fan Wheel shall be backward-curved non-over loading type. Fan wheel and housing shall be statically and dynamically balanced. For fans upto 450 mm dia, fan outlet velocity shall not exceed 550 meter/minute and maximum fan speed shall not exceed 1450 rpm. For fans above 450 mm dia, the outlet velocity shall be within 700 meter/minute and maximum fan speed shall not exceed 1000 RPM. High static pressure fan speed shall beas per manufacturer.

Shaft shall be constructed of steel, turned, ground and polished.

Bearings : shall be of the sleeve / ball-bearing type mounted directly on the fan housing. Bearings shall be designed especially for quiet operation and shall be of the self-aligning, oil / grease pack pillow block type.

Motor : Fan motor shall be energy efficient and suitable for 415±10% volts, 50 cycles, 3 phase AC power supply, squirrel-cage, totally enclosed, fan-cooled motor, provided with class F insulation and meeting IE-2 efficiency.

Drive to fan shall be provided through belt with adjustable motor sheave and a standard belt guard. Belts shall be of the oil-resistant type.

Vibration Isolation : MS base shall be provided for both fan and motor, built as an integral part, and shall be mounted on a concrete foundation through resistoflex vibration isolators. The concrete foundation shall be atleast 15 cm above the finishedfloor level, or as shown in approved-for-construction shop drawings.

Centrifugal fans for smoke extract application shall have external belt drive and motor. Fan & casing shall be internally rated for 300$^{o}$C for 2 hours.

### 3.2 Mixed Flow Inline Fan

Circular Mixed Flow Inline fan shall incorporate dual speed Mixed Flow direct driven ABS Impeller for high pressure & low noise level. The fan assembly shall be encased in shockproof & corrosion resistant casing along with connection box. The fan casing shall be of High Grade plastic material.
The Fan can be easily mounted on ceiling and mounting frame shall be provided to facilitate easy connection and access to the fan/impeller without dismantling the duct for maintenance & cleaning purpose. Flexible anti-vibration joints shall be provided to arrest

vibration being transformed to other equipment connected to inline fan. Motor shall be dualspeed, single phase having IP44 protection with sealed ball bearing.
The Fan should be installed along with wired plug and 2 speed controller should be wired inconnection box for easy electrical connection at site.

### 3.3 Axial Flow Fan

Fan shall be complete with motor, motor mount, belt driven (or direct driven) andvibration isolation type, suspension arrangement as per approved for construction shop drawings.

**Casing** : shall be constructed of heavy gage sheet steel. Fan casing, motor mount and straightening vane shall be of welded steel construction. Motor mounting plate shall be minimum 15 mm thick and machined to receive motor flange.

An inspection door with handle and neoprene gasket shall be provided. Casing shall have flanged connection on both end for ducted applications. Fan casing are with internal punched inlet and outlet flanges to prevent air leakage, for size upto 1600 mm dia and shall be constructed of rolled steel with a continuous seam welded. Support brackets for ceiling suspension shall be welded to the casing for connection to hanger bolts. Straightening vanes shall be aerodynamically designed for maximum efficiency by converting velocity pressure to static pressure potential and minimizing turbulence. Casing shall be bonderized, primed (minimum 2 coats of rust-proof primer) and finish coated with enamel paint or powder coated after phosphating process.

**Rotor :** hub and blades shall be cast aluminium alloy or cast steel construction. Blades shall be die-formed aerofoil shaped for maximum efficiency and shall vary in twist and width from hub to tip to effect equal air distribution along the blade length. Rotor shall be statically and dynamically balanced. Extended grease leads for external lubrication shall be provided. The fan pitch control may be manually readjusted at site upon installation, for obtaining actual air flow values, as specified and quoted. Taper lock bushing shall be used to mount the propeller to the motor shaft. The impeller and fan casing shall be carefully matched and shall have precise running tolerances for maximum performance and operating efficiency.

**Motor**: shall be energy efficient squirrel-cage, totally-enclosed, fan cooled, standard frame, constant speed, continuous duty, single winding, suitable for 415±10% volts, 50 cycles, 3 phase AC power supply, provided with class ‘ F’ insulation. Motor shall be specially designed for quiet operation. The speed of the fans shall not exceed 1000 RPM for fans with impeller diameter above 450 mm, and 1440 RPM for fans with impeller diameter 450 mm and less. For lowest sound level, fan shall be selected for maximum efficiency or minimum horsepower. Motor conduit box shall be mounted on exterior of fan casing, and lead wires from the motor to the conduit box shall be protectedfrom the air stream by enclosing in a flexible metal conduit.

**Drive** : Fan shall be provided through direct drive

**Vibration Isolation** : The assembly of fan and motor shall be suspended from the slab by vibration isolation suspension of heavy duty spring isolators type.

*Axial Flow Fan shall be AMCA certified for Air and Sound performance in accordance to AMCA 210 and AMCA 300. Fan shall be suitable for both indoor and outdoor application with all accessories. Base fan performance shall be at standard conditions (density 1.2 Kg/Cu.mt.)*

*Following table shall be followed while selecting the fan :*

| S.No | Description | Standard Fan | UL Listed Fan | Fire Rated Fan |
|---|---|---|---|---|
| 1 | Casing | It shall be constructed of heavygauge sheet steel . | It shall be constructed of heavy gauge sheet steel and should come under ULStandards. | It shall be constructed of heavy gauge sheet steel and shall withstand 300 degree Cfor 2 hours & should be fire rated. |
| 2 | Rotor | Hub & Blades shallbe cast aluminum alloy or cast steel construction . | Hub & Blades shall be cast aluminum alloy or cast steel construction and should come under UL Standards. | Hub & Blades shall be cast aluminum alloy or cast steel construction and shall withstand 300 degree C for 2 hours & should be fire rated. |
| 3 | Motor | Motor shall have class F insulation. | Motor shall pass elevated temperature test and othertests as per UL Standards and shall be UL listed. | Motor for emergency fire, smoke and heat ventilation shall certified according to standard BS EN 12101-3:2002 for 300 degree C for 2 hours & should be fire rated. |

## SPLIT AIR CONDITIONING UNITS

### 4.0 Scope

The scope of this section comprise the supply, erection, testing and commissioning of Split Units conforming to these specifications and in accordance with the requirements of Drawings and Schedule of Quantities.

### 4.1 Type

The Units shall consist of hermetically sealed Scroll compressor, motor, air cooled condenser, sump heaters, integral refrigerant piping and wiring, all mounted on a steel frame.

Indoor unit to be installed for Split Unit within building, shall be housed in insulated cabinet consisting of cooling coil, blower with motor, filter & insulated drain pan.

### 4.2 Capacity

The refrigeration capacity split unit shall be as shown on Drawings and in Schedule of Quantities.

### 4.3 Compressor And Motor

Compressor shall be hermetic Scroll, serviceable type and shall have dual pressure stat, and an operating oil charge. The motor shall be suction gas cooled and shall be sealed against dirt and moisture. The motor shall be suitable for 415±10% volts or 230±10 % volts, 50 Hz, A.C. supply.

### 4.4 Refrigerant Piping and Controls

Refrigerant piping and fittings interconnecting compressor and condenser shall be all copper and valves shall be brass / gunmetal construction.

### 4.5 Casing

The indoor & outdoor units shall be sectionalised / cabinet construction. Indoor units shall be consisting of fan section, coil section, filter section, and drain pan. Outdoor unit shall consist of condenser coil, fan & compressor. In case of split unit, the compressor shall be mounted with the outdoor units. Each section shall be constructed of thick sheet steel all welded / bolted construction, adequately reinforced with structural members and provided with sufficient access panels for proper lubrication and maintenance. Base panel shall be constructed of fabricated steel structure provided with an under frame suitably braced. Each unit shall include one piece drain pan constructed of 20 gauge galvanised sheet steel plate. Drain pan shall extend under coil and fan sections with drain connections. Removable panels in fan and coil sections shall provide access to all internal parts. Panels shall be internally lined with 2.5 cm thick fibre glass as per section “Insulation” for the thermal insulation and acoustic lining.

### 4.6 Fan Motor And Drive

Fan motor shall be suitable for 415 ± 10% volts or 230±10% volts, 50 Hz, A.C. Supply, Single phase, motors shall be provided with permanent capacitor. Motors shall be especially designed for quite operation and motor speed shall not exceed 1440 rpm

### 4.7 Fan

Fan wheels and housing shall be fabricated from heavy gauge steel. Fan wheels shall be of double-width, double inlet forward-curve, multi-blade type enclosed in a housing and mounted on a common shaft. Fan housing shall be made of die-formed steel sheets with stream-lined inlets to ensure smooth air flow into the fans, fan shaft bearing shall be oil/grease lubricated. All rotating parts shall be dynamically balanced individually, and the complete assembly shall be statically and hydraulically balanced. Fan speed shall not exceed 1000 rpm and maximum fan outlet velocity shall be 550 meters per minute.

### 4.8 Cooling Coil

Cooling coils shall be of fin and tube type having aluminium fins firmly bonded to copper tubes assembled in zinc coated steel frame. Face and surface areas shall be such as to ensure rated capacity from each unit and air velocity across each coil shall not exceed 100 meters per minute. The coil shall be pitched in the unit casing for proper drainage. Each coil shall be factory-tested at 21 Kg. per sq.cm air pressure under water. Tube shall be mechanically / hydraulically expanded for minimum thermal contract resistance with fins. The number of fins shall be 4 to 5 per cm..

### 4.9 Vibration Isolators

The indoor and outdoor units shall be provided with ribbed rubber pad vibration isolators.

### 4.10 Painting

Units shall be factory finished with durable alkyd spray enamel. Shop coats of paint that have become marred during shipment or erection shall be cleaned off with mineral spirits, then coated with enamel paint to match the finish over the adjoining shop-painted surface.

### 4.11 Performance Rating

The unit shall be selected for the lowest operating noise level. Capacity rating and power consumption with operating points clearly indicated shall be submitted with the tenders and verified at the time of testing and commissioning of the installation.

## 5.0 PIPING

### 5.1 Scope

The scope of this section comprises the supply and laying of pipes, pipe fittings and valves, testing and balancing of all condensate and refrigerant piping required for the complete installation as shown on the Drawings. All piping inclusive of fittings and valves shall follow the applicable Indian Standards.

### 5.2 Pipe Sizes

Pipe sizes shall be as required for the individual fluid flows. Various pipe sizes have been indicated on the Drawings, these are for Contractor's guidance only and shall not relieve contractor of responsibility for providing smooth noiseless balanced circulation of fluids.

### 5.3 Drain Piping

a. The piping system shall consist of Non-pressure pipe up to 10 kg/ sq.mm UPVC piping from 15 mm to 50 mm .

b. For any internal works, the UPVC pipes and fittings shall be embedded in the wall chase or run on the floor/ceiling unless otherwise specified. No unsightly exposed runs shall be permitted.

c. For proper drainage of Condensate, 'U' trap shall be provided in the drain piping.

d. All condensate drain piping shall be insulated and painted as per the section "Insulation" as indicated in Schedule of Quantities.

### 5.4 Refrigerant Piping

a. All refrigerant pipes and fittings shall be 18 G hard drawn copper tubes and wrought copper / brass fittings suitable for connection with silver solder / phos-copper.

b. All joints in copper piping shall be sweat joints using low temperature brazing and / or silver solder. Before joining any copper pipe or fittings, its interiors shall be thoroughly cleaned by passing a clean cloth via wire or cable through its entire length. The piping shall be continuously kept clean of dirt etc. while constructing the joints. Subsequently, it shall be thoroughly blown out using carbon dioxide / nitrogen. Refrigerant lines shall be sized to limit pressure drop between the evaporator and condensing unit to less than 0.2 kg per sq.cm.

c. Sight glass with moisture indicator and removable type combination dryer cum filter with MS housing and brass wire mesh / punched brass sheet shall be installed in liquid line of the refrigeration system incorporating a three valve by pass. After ninety days of operation, liquid line drier cartridges shall be replaced.

d. Heat exchanger shall be MS heavy duty pipe in pipe type and without any joint in the inner pipe. Horizontal suction line shall be pitched towards the compressor and no reducers shall be provided for proper oil return.

g. After the refrigerant piping installation has been completed, the refrigerant piping system shall be pressure tested using Freon mixed with nitrogen / carbon dioxide at a pressure of 20 kg per sq. cm (high side) and 10 kg per sq. cm (low side). Pressure shall be maintained in the system for a minimum of 12 hours. The system shall then be evacuated to a minimum vacuum of 70 cm of mercury and held for 24 hours. Vacuum shall be checked with a vacuum gage. All refrigeration piping shall be installed strictly as per the instructions and recommendations of air conditioning equipment manufacturer.

## 6.0 AIR DISTRIBUTION

### 6.1 Scope

The scope of this section comprises supply fabrication, installation and testing of all sheet metal / aluminum ducts, supply, installation, testing and balancing of all grilles, registers and diffusers. All to be in accordance with these specifications and the general arrangement shown on the Drawings.

### 6.2 Duct Material

Raw Materials

Galvanized steel sheets with Class - VIII Galvanizing – light coating of zinc (Zinc coating shall be Lead free), nominal 120gm/sq.m surface area and Lock Forming Quality prime material along with mill test certificates. In addition, if deemed necessary, samples of raw material, selected at random by owner's site representative shall be subject to approval and tested for thickness and zinc coating at contractor's expense.

Gauges, Bracing By Size Of Ducts

All ducts shall be factory fabricated from galvanized steel of the following thickness, as indicated below :

| Rectangular Ducts G. S. | External Pressure 250 Pa | | |
|---|---|---|---|
| | Duct Section Length 1.2 m (4 ft) | | |
| **Maximum Duct Size** | **Gauge** | **Joint Type** | **Bracing Spacing** |
| 1–500 mm | 24 | C&S Connector | Nil |
| 501 – 750 mm | 24 | C&S Connector | Nil |
| 751 – 900 mm | 24 | TDF Flange | Nil |
| 901 – 1200 mm | 24 | TDF Flange | Nil |
| 1201 – 1500 mm | 22 | TDF Flange | Nil |
| 1501 – 1800 mm | 22 | TDF Flange | JTR or ZEE BAR |
| 1801 – 2100 mm | 20 | TDF Flange | JTR or ZEE BAR |
| 2101 – above | 18 | TDF Flange | JTR or ZEE BAR |

*Distance of reinforcement/bracing from each joint. Bracing material to be same as of material used for joining of duct sections.

### 6.3 FABRICATION STANDARDS & EQUIPMENT

All duct construction and installation shall be in accordance with SMACNA standards. In addition ducts shall be factory fabricated utilizing the following machines to provide the requisite quality of ducts.

1. Coil (Sheet metal in Roll Form) lines to facilitate location of longitudinal seams at corners/folded edges only, for required duct rigidity and leakage free characteristics. No longitudinal seams permitted along any face side of the duct.

2. All ducts, transformation pieces and fittings to be made on CNC profile cutter for requisite accuracy of dimensions, location and dimensions of notches at the folding lines.

3. All edges to be machine treated using lock formers, flangers and rollers for turning up edges

.

## 6.4 DUCT CONSTRUCTION

All ducts shall be fabricated and installed in workmanlike manner, conforming to relevant SMACNA codes.

a) Ducts so identified on the Drawings shall be acoustically lined and insulated from outside as described in the section “Insulation” and as indicated in schedule of Quantities. Duct dimensions shown on drawings, are overall sheet metal dimensions inclusive of the acoustic lining where required and indicated in Schedule of quantities. The fabricated duct dimensions should be as per approved drawings and care should be taken to ensure that all connecting sections are dimensionally matched to avoid any gaps.

b) Ducts shall be straight and smooth on the inside with longitudinal seams shall be airtight and at corners only which shall be either Pittsburgh or snap button as per SMACNA practice, to ensure air tightness.

c) All ducts up to 75cms width within conditioned spaces shall have C&S connector. The internal ends of slip joints shall be in the direction of airflow. Care should be taken to ensure that Cleats are mounted on the longer side of the duct and Cleats on the shorter side. Ducts and accessories within ceiling spaces, visible from air-conditioned areas shall be provided with two coats of mat black finish paint.

d) Changes in dimensions and shape of ducts shall be gradual (between 1:4 and 1:7). Air-turns (vanes) shall be installed in all bends and duct collars designed to permit the air to make the turn without appreciable turbulence.

e) Ducts shall be fabricated as per details shown on Drawings. All ducts shall be rigid and shall be adequately supported and braced where required with standing seams, tees, or angles, of ample size to keep the ducts true to shape and to prevent buckling, vibration or breathing.

f) All sheet metal connection, partitions and plenums, required to confine the flow of air to and through the filters and fans, shall be constructed of 18 gauge GSS / 16gauge aluminum, thoroughly stiffened with 25mm x 25mm x 3mm galvanized steel angle braces and fitted with all necessary inspection doors as required, to give access to all parts of the apparatus. Access doors shall be not less than 45cm x 45cm in size.

g) Plenums shall be shop/factory fabricated panel type and assembled at site. Fixing of galvanized angle flanges on duct pieces shall be with rivets heads inside i.e. towards GS sheet and riveting shall be done from outside.

h) Self adhesive Neoprene rubber / UV resistant PVC foam lining 5mm nominal thickness instead of felt, shall be used between duct flanges and between duct supports in all ducting installation

### 6.5 INSTALLATION PRACTICE

All ducts shall be installed generally as per tender drawings, and in strict accordance with approved shop drawings to be prepared by the Contractor:

a) The Contractor shall provide and neatly erect all sheet metal work as may be required to carry out the intent of these Specifications and Drawings. The work shall meet with the approval of Owner's site representative in all its parts and details

b) All necessary allowances and provisions shall be made by the Contractor for beams, pipes, or other obstructions in the building, whether or not the same are shown on the drawings. Where necessary to avoid beams or other structural work, plumbing or other pipes, and conduits, the ducts shall be transformed, divided or curved to one side (the required area being maintained) all as per the site requirements.

c) If a duct cannot be run as shown on the drawings, the contractor shall install the duct between the required points by any path available in accordance with other services and as per approval of owner's site representative.

d) All ductwork shall be independently supported from building construction. All horizontal ducts shall be rigidly and securely supported, in an approved manner, with hangers formed of galvanized steel wire ropes (as per clause 20.12) and galvanized steel angle/channel or a pair of brackets, connected by galvanized steel wire hangers under ducts, rigid supports may be provided at certain interval if need be. The spacing between supports should be not greater than 2.4 meter. All vertical ductwork shall be supported by structural members on each floor slab. Duct supports may be through galvanized steel insert plates or Toggle end wire fixing left in slab at the time of slab casting.

e)

Galvanized steel cleat with a hole for passing the wire rope hanger shall be welded to the plates. Trapeze hanger formed of galvanized steel wire rope using Gripple shall be hung through these cleats. Wherever use of metal insert plates is not feasible, duct support shall be through dash/anchor fastener driven into the concrete slab by electrically operated gun. Wire rope supports shall hang through the cleats or wire rope threaded studs can be screwed into the anchor fasteners.

Alternatively, if mentioned in the SOQ, all ductwork shall be independently supported from building construction. All horizontal ducts shall be rigidly and securely supported, in an approved manner, with trapeze hangers formed of galvanized steel rods and galvanized steel angle/channel or a pair of brackets, connected by galvanized steel rod under ducts**.** The spacing between supports should be not greater than 2.0 meter. All vertical ductwork shall be supported by structural members on each floor slab. Duct supports may be through galvanized steel insert plates left in slab at the time of slab casting. Galvanized steel cleat with a hole for passing the hanger rods shall be welded to the plates. Trapeze hanger formed of galvanized steel rods shall be hung through these cleats. Wherever use of metal insert plates is not feasible, duct support shall be through dash/anchor fastener driven into the concrete slab by electrically operated gun. Hanger rods shall then hang through the cleats or fully threaded galvanized rods can be screwed into the anchor fasteners.

Ducting over furred ceiling shall be supported from the slab above, or from beams after obtaining approval of Owner's site representative. In no case shall any duct be supported from false ceiling hangers or be permitted to rest on false ceiling. All metal work in dead or furred down spaces shall be erected in time to occasion no delay to other contractor's work in the building.

Where ducts pass through brick or masonry openings, it shall be provided with 25mm thick TF quality expanded polystyrene around the duct and totally covered with fire barrier mortar for complete sealing.

All ducts shall be totally free from vibration under all conditions of operation. Whenever ductwork is connected to fans, air handling units or blower coil units that may cause vibration in the ducts, ducts shall be provided with a flexible connection, located at the unit discharge. Flexible connections shall be constructed of fire retarding flexible heavy canvas sleeve at least 10cm long securely bonded and bolted on both sides. Sleeve shall be made smooth and the connecting ductwork rigidly held by independent supports on both sides of the flexible connection. The flexible connection shall be suitable for pressure at the point of installation.

Duct shall not rest on false ceiling and shall be in level from bottom. Taper pieces shall taper from top.

## 6.6 DAMPERS

a. Dampers: All duct dampers shall be opposed blade louver dampers of robust 16 G GSS construction and tight fitting. The design, method of handling and control shall be suitable for the location and service required.

b. Dampers shall be provided with suitable links levers and quadrants as required for their proper operation. Control or setting device shall be made robust, easily operable and accessible through suitable access door in the duct. Every damper shall have an indicating device clearly showing the damper position at all times.

c. Dampers shall be placed in ducts at every branch supply or return air duct connection, whether or not indicated on the Drawings, for the proper volume control and balancing of the air distribution system.

d. Pressure relief dampers: Pressure relief dampers shall be constructed with 18G Aluminum construction with parallel blade construction. Leafs shall be 100% air tight upon closure. Leafs shall be loaded with spring pressure of stiffness (k value) corresponding to set point pressure.

e. Non return damper (Back draft damper) : Non return damper shall be constructed out of 16G GSS. Blades shall ensure 100% air leak proof performance on closure. Design shall ensure that no rattling noise is produced at design duty.

f. Constant Volume Regulator (For Hotel/ Hospital TFA / Exhaust ducts)

Constant volume regulators (KVR) shall be used to obtain constant air volume at a given pressure range.

The constant volume regulators (KVR) shall be of the circular type for high pressures and to be inserted into ductwork and suitable for vertical as well as horizontal mounting and it should be placed at a minimum distance of 3x the duct diameter from air supply grilles and minimum distance of 1x the duct diameter from air exhaust grilles

Constant volume regulator body, valve and piston shall be made out of flame retardant PVC, fire classification M1. They shall contain a self regulating PVC valve, piston, rubber strip for air tightness inside the duct and stainless steel calibrated spring and shall have preset air volume.

## 6.7 FIRE & SMOKE DAMPERS

a. All supply and return air ducts at AHU room crossings and at all floor crossings or as indicated in the drawings shall be provided with Motor operated Fire & smoke damper of at least 90 minutes rating. These shall be of multi-leaf type and provided with Spring Return electrical actuator having its own thermal trip for ambient air temperature outside the duct and air temperature inside the duct. Actuator shall have Form fit type of mounting, metal enclosure and guaranteed long life span. The dampers shall meet the requirements of NFPA90A, 92A and 92B. Dampers shall have a fire rating of 1.5 Hrs. in accordance with latest edition of UL555 and shall be classified as Leakage Class 2 smoke damper in accordance with latest version of UL555S. Each fire/smoke damper shall be AMCA licensed and bear the AMCA seal for air Performance. Pressure drop shall not exceed 7.5Pa when tested at 300m/min face velocity on 600x600mm size damper. Actuator shall be UL listed.

b. Each damper shall be supplied with factory mounted sleeve of galvanized steel of thickness as per SMACNA and of minimum 500mm long or as specified in schedule of quantities depending up on the wall thickness. The damper shall be fitted in to sleeve either using welding or self tapping screws. All welded joints shall be finished using heat resistance steel paint . UL listed and approved Silicon sealant shall be applied at all corners as well as at joints between damper frame and sleeve. Damper Frame shall be a roll formed structural hat channel , reinforced at corners, formed from a single piece of 1.6mm galvanized steel . Damper blades shall be airfoil shaped (equivalent to 2.3mm thickness strength) roll formed using 0.8mm thick single piece of galvanized sheet. Bearings shall be of stainless steel fitted in an extruded hole in the damper frame. Blade edge seals shall be silicone rubber and galvanized steel mechanically locked in to the blade edge (adhesive type seals are not acceptable). Side Jam seals of stainless steel and Top and bottom seals of galvanized steel shall be provided. All galvanized steel used shall be with minimum 180 gm / sqm Zinc coating. Bigger size Dampers shall be supplied in Multiple modules of sizes not exceeding in dimensions of certified module, jack shafted together. Multiple actuators shall be provided for large dampers with higher torque requirements as prescribed in UL.

c. The electric actuator shall be energized either upon receiving a signal from smoke detector installed in AHU room supply air duct / return air duct. Electric Actuator of suitable Torque and as approved by UL shall be factory mounted and tested. The actuator shall be suitable for 24V AC supply.In addition actuator shall have elevated temperature rating of 250 deg.F. Electric Actuator shall have been energized hold open tested for a period of at least one year with no spring return failure. Each fire/smoke damper shall be equipped with a heat actuated release device which shall allow controlled closure of damper rather than instantaneous to prevent accident.( Electrical fusible link).The EFL shall allow the damper to reopen automatically after a test, smoke detection or power failure condition. The damper shall be equipped with a device to indicate OPEN and CLOSE position of Damper blades through a link mounted on the damper blade.

d. Each damper shall be provided with its own control panel, mounted on the wall and suitable for 240 VAC supply. This control panel shall be suitable for spring return actuator and shall have atleast the following features:

- Potential free contacts for AHU fan ON/ Off and remote alarm indication.
- Accept signal from external smoke / fire detection system for tripping the electrical actuator.

- Test and reset facility.
- Indicating lights / contacts to indicate the following status:
- Power Supply On
- Alarm
- Damper open and close position.

e. Actuators shall be mounted on the sleeve by the damper supplier in his shop and shall furnish test certificate for satisfactory operation of each Motor Operated Damper in conjunction with it's control panel. Control panel shall be wall mounted type.

f. It shall be HVAC Contractor's responsibility to co-ordinate with the Fire Alarm System Contractor for correctly hooking up the Motor Operated Damper to Fire Detection / Fire Management System. All necessary materials for hooking up shall be supplied and installed by HVAC Contractor under close co-ordination with the fire protection system contractor.

g. HVAC Contractor shall demonstrate the testing of all Dampers and its control panel after necessary hook up with the fire protection / fire management system is carried out by energizing all the smoke detectors with the help of smoke.

h. HVAC Contractor shall provide Fire retardant cables wherever required for satisfactory operation and control of the Damper.

j. HVAC Contractor shall strictly follow the instructions of the Damper Supplier or avail his services at site before carrying out testing and installation at site.

k. Fire/smoke damper shall be provided with factory fitted sleeves; however, access doors shall be provided in the ducts within AHU room in accordance with the manufacturer's recommendations.

l. The Contractor shall also furnish to the Owner, the necessary additional spare actuators and temperature sensor (a minimum of 5% of the total number installed) at the time of commissioning of the installation.

**6.8 FIRE DAMPERS**

a. Whenever a supply/return duct crosses from one fire zone to another, it shall be provided with approved fire damper of at least 1½ hour fire rating as per UL555/1995 tested by CBRI. This shall be curtain type fire damper.

b. Fire damper blades shall be one piece folded high strength 16 gage galvanized steel construction. In normal position, these blades shall be gathered and stacked at the frame head providing maximum air passage and preventing passing air currents from creating noise or chatter. The blades shall be held in position through fusible link of temp 74oC (165°F).
The HVAC contractor shall supply UL classified Fire Dampers meeting or exceeding the specifications. Fire Dampers shall be furnished and installed at locations shown in Drawings and as described in Schedule of quantities. Fire Dampers shall have a fire rating of 1.5/3 Hrs.as specified in BOQ, in accordance with latest edition of UL555. Each Fire damper shall be AMCA licenced and shall bear the AMCA seal for air performance.

Damper shall be equipped with UL labelled Fusible Link with Temperature setting 74°C (165°F) or as specified in Schedule of quantities. Fire dampers shall have been tested to close under dynamic air flow conditions with pressure up to 1000 pa and velocities up to 10.2 m /sec. Fire damper shall be approved for Horizontal or vertical installation as may be required by the location shown in the drawings.

Damper Frame shall be a roll formed structural hat channel, reinforced at corners, formed from a single piece of 1.6mm galvanized steel. Damper blades shall be roll formed 3-v groove (1.6mm thick) or airfoil shaped in case of 3 Hrs. fire rating (equivalent to 2.3mm thickness strength) roll formed using 0.8mm thick single piece of galvanized sheet. Bearings shall be of stainless steel fitted in an extruded hole in the damper frame. All galvanized steel used shall be with minimum 180 gm / sqm Zinc coating Bigger size Dampers shall be supplied in Multiple modules of sizes not exceeding in dimensions of certified module jack shafted together.

Fire damper shall be equipped with a electric limit switch to indicate open and close position of the damper blades.

Fire Damper shall be installed in wall or floor opening using galvanized steel sleeve of minimum 500mm length of sheet thickness for fire & smoke damper as per SMACNA and as per Installation instruction of Manufacturer.

c. In case of fire, the intrinsic energy of the folded blades shall be utilized to close the opening. The thrust of the suddenly released tension shall instantly drive the blades down and keep it down without the use of springs, weights or other devices subject to failure.

d. Fire damper sleeves and access doors shall be provided within the duct in accordance with the manufacturer's recommendation.

e. The contractor shall also furnish to the Owner, the necessary additional fusible links (spares), minimum of 5% of the total number installed, at the time of commissioning of the installation.

## 6.9 SUPPLY AND RETURN AIR REGISTERS

Supply & return air registers shall be of either steel or aluminium sections as specified in schedule of quantities. Steel construction registers shall have primer Coat finish whereas extruded aluminium registers shall be either Anodised or Powder Coated as specified in Schedule of Quantities. These registers shall have individually adjustable louvers both horizontal and vertical. Supply air registers shall be provided with key operated opposed blade extruded aluminium volume control damper anodised in matt black shade.

The registers shall be suitable for fixing arrangement having concealed screws as approved by Architect. Linear continuous supply cum return air register shall be extruded aluminium construction with fixed horizontal bars at 15 Deg. inclination & flange on both sides only (none on top & bottom). The thickness of the fixed bar louvers shall be minimum 5.5 mm in front and 3.8 mm in rear with rounded edges. Flanges on the two sides shall be 20 mm/30 mm wide as approved by Architect.

The grilles shall be suitable for concealed fixing. Volume control dampers of extruded aluminium anodised in black color shall be provided in supply air duct collars. For fan coil units horizontal fixed bar grilles as described above shall be provided with flanges on four sides, and the core shall be & suitable for clip fixing, permitting its removal without disturbing the flanges.

a. All registers shall be selected in consultation with the Architect. Different spaces shall require horizontal or vertical face bars, and different width of margin frames. These shall be procured only after obtaining written approval from Architect for each type of register.

b. All registers shall have a soft continuous rubber/foam gasket between the periphery of the register and the surface on which it has to be mounted. The effective area of the registers for air flow shall not be less than 66 percent of gross face area.

c. Registers specified with individually adjustable bars shall have adjustable pattern as each grille bar shall be pivotable to provide pattern with 0 to +45 degree horizontal arc and upto 30 degree deflection downwards. Bars shall hold deflection settings under all conditions of velocity and pressure.

d. Bar longer than 45 cm shall be reinforced by set-back vertical members of approved thickness.

e. All volume control dampers shall be anodised aluminium in mat black shade.

## 6.10 SUPPLY AND RETURN AIR DIFFUSERS

Supply and return air diffusers shall be as shown on the Drawings and indicated in Schedule of Quantities. Mild steel diffusers/dampers shall be factory coated with rust-resistant primer. Aluminium diffusers shall be powder coated & made from extruded aluminium section as specified in schedule of quantities.

a. Rectangular Diffusers shall be steel / extruded aluminium construction, square & rectangular diffusers with flush fixed pattern for different spaces as per schedule of quantities These shall be selected in consultation with the Architect. These shall be procured only after obtaining written approval from Architect for each type of diffuser.

b. Supply air diffusers shall be equipped with fixed air distribution grids, removable key-operated volume control dampers, and anti-smudge rings as re-required in specific applications and as per requirements of schedule of quantities. All extruded aluminium diffusers shall be provided with removable central core and concealed key operation for volume control damper.

c. Linear Diffuser shall be extruded aluminium construction with removable core, one or two way blow type. Supply air diffusers shall be provided with volume control/ balancing dampers within the supply air collar. Diffusers for different spaces shall be selected in consultation with the Architect, and provided as per requirements of schedule of quantities. All diffusers shall have volume control dampers of extruded aluminium construction anodised in mat black shade.

d. Slot Diffuser shall be extruded aluminium construction multislot type with air pattern controller provided in each slot. Supply air diffusers shall be provided with Hit & Miss volume control dampers in each slot of the supply air diffusers. Diffusers for different spaces shall be selected in consultation with the Architect and provided as per requirement of Schedule of Quantities.

e. Data centers shall be provided with floor grilles. Grilles shall be of nominal size of 600mm x 600mm and shall be fitted in floor tile of false floor. Grille shall be with dampers for flow control. Grill shall be heavy duty 16G Aluminium and shall take care of human traffic load. Damper shall be operable in situ without requirement of removal of grille.

## 6.11 HVAC Supports

Gripple Hanger Supports are suitable for: Rectangular duct, Spiral Duct, Oval Duct, Fabric Duct, Diffusers, plenum boxes

6.11.1 Ducting Supports:

All ductwork shall be independently supported from building construction. All horizontal ducts shall be rigidly and securely supported, in an approved manner, with hangers formed of galvanized steel wire ropes and galvanized steel angle/channel or a pair of brackets, connected by galvanized steel wire hangers under ducts, rigid supports may be provided at certain interval if need be. The spacing between supports should be not greater than 2.4 meter. All vertical ductwork shall be supported by structural members on each floor slab. Duct supports may be through galvanized steel insert plates or Toggle end wire fixing left in slab at the time of slab casting. Galvanized steel cleat with a hole for passing the wire rope hanger shall be welded to the plates. Trapeze hanger formed of galvanized steel wire rope using Gripple shall be hung through these cleats. Wherever use of metal insert plates is not feasible, duct support shall be through dash/anchor fastener driven into the concrete slab by electrically operated gun. Wire rope supports shall hang through the cleats or wire rope threaded studs can be screwed into the anchor fasteners. In case of PEB structure Loop and Catenary system can be used based on the site conditions as per approved suspension system drawings.

All horizontal ducts shall be adequately secured and supported. In an approved manner, with trapeze Hangers formed of galvanized steel wire rope in a cradle support method (refer to typical drawings) under ducts at no greater than 3000mm centre, for 3001mm-above appropriate size angle along with neoprene pad in between the duct & MS angle should be used with prior approval. All vertical duct work shall be supported by structural members on each floor slab. Duct support shall be through dash / anchor fastener driven into the concrete slab by electrically operated gun. Hanger wires shall then hang around the ducting. Rigid supports shall be used in conjunction with wire rope hangers to assist with alignment of services where recommended for by the manufacturer. Rigid support must also be used in conjunction with wire rope hangers with duct work at each change of direction or connection or as per approved drawings. Support ducting in accordance with Schedule I specified below. Any other Gripple solution can be used based on manufacturer's recommendation on site conditions after prior approval. In cases of Spiral ducting the wire can be wrapped directly around the ducting without the need for a spiral ducting clamp for sizes above 1100 a cradle support should be provided, refer to manufacturer's recommendations.

Ducting over furred ceiling shall be supported from the slab above or from beams after obtaining approval of Construction manager/consultant. In no case shall any duct be supported from false ceiling Hangers or be permitted to rest on false ceiling. All metal work in dead or furred down spaces shall be erected in time to occasion no delay to other Contractor's work in the building. All supports of pipe shall be taken from structural slab/wall by means of fastener.

**Catenary Supports**: Refer to manufacturer's recommendations on Catenary supports with C-clip, special care should be taken with tensioning of the wire and angles at which the installation of services are made.

Stainless Steel Supports should be provided for food, chemical and High Corrosion areas near coastlines.

For further technical information refer to manufacturers catalogue and installation guide. **Comply with manufacturer's load ratings and recommended installation procedures.**

**Schedule I: Duct Hanger Schedule**

| For ducts with external SP upto 250 Pa | | | For ducts with external SP upto 500 Pa | | |
|---|---|---|---|---|---|
| **Maximum Duct Size (mm)** | **Gauge** | **Gripple Hanger size** | **Maximum Duct Size (mm)** | **Gauge** | **Gripple Hanger size** |
| 1 - 500 | 24 | No. 1 or 2 | 1–400 mm | 24 | No. 2 |
| 501 - 750 | 24 | No. 1 or 2 | 401-700 mm | 24 | No. 2 or 3 |
| 751 - 900 | 24 | No. 2 | 701-900 mm | 22 | No. 2 or 3 |
| 01 - 1200 | 24 | No. 2 or 3 | 901-1000 mm | 22 | No. 3 or 4 |
| 201 - 1500 | 22 | No. 3 | 1001-1200 mm | 22 | No. 3 or 4 |
| 501 - 1800 | 22 | No. 3 or 4 | 1201-2100 mm | 18 | No. 3 or 4 |
| 801-2100 | 20 | No. 3 or 4 | 2101 - 3000mm | 18 | No. 4 |
| 101-3000 | 18 | No. 4 | 3001 - above (Trapeze type support Arrangement) | 18 | No. 3 or 4 |
| 3001 - above (Trapeze type support Arrangement) | 18 | No. 3 or 4 | | | |

**Notes: All supports are considered at 2400 mm interval in above table and may vary as per the design but should not be greater than 2400mm.**

All units shall be adequately secured and supported in an approved manner using wire hanger suspension Y fit solution as per manufacturers' recommendation with prior approval.

**Rigid Supports to be used in conjunction with wire supports:**

Rigid supports if required in conjunction with wire hangers shall be of steel, adjustable for height and Zinc chromate primer coated and finish coated black. Where supports and clamps are of dissimilar materials, a gasket shall be provided in between. If the MS angle at the bottom if required as per design should be as per following table:

| **Longer size of Duct (mm)** | **Type of Joints** |
|---|---|
| Up to 750 | 25x25x3 mm L angle with M8 nuts & bolts |
| 751-1000 | 25x25x3 mm L angle with M8 nuts & bolts |
| 1001-1500 | 40x40x5 mm L angle with M8 nuts & bolts |
| 1501-2250 | 50x50x5 mm L angle with M10 nuts & bolts |
| 2251 & above | 50x50x6 mm L angle with M10 nuts & bolts |

All the supporting system should be supplied from the same manufacturer.

## 6.12 DOCUMENTATION & MEASUREMENTS FOR DUCTING

All ducts fabricated and installed should be accompanied and supported by proper documentation viz:

a) Bill of material/Packing list for every duct section supplied.

Measurement sheet covering each fabricated duct piece showing dimensions and external surface area along with summary of external surface area of duct gauge-wise.

Each and every duct piece to have a tag number, which should correspond to the serial number, assigned to it in the measurement sheet. The above system will ensure speedy and proper site measurement and verification.

Unless otherwise specified, measurements for ducting for the project shall be on the basis of centerline measurements described herewith:

Ductwork shall be measured on the basis of external surface area of ducts. Duct measurements shall be taken before application of the insulation. The external surface area shall be calculated by measuring the perimeter comprising overall width and depth, including the corner joints, in the center of each duct section, multiplying with the overall length from flange face to flange face of each duct section and adding up areas of all duct sections. Plenums shall also be measured in a similar manner.

For tapered rectangular ducts, the average width and depth shall be considered for perimeter, whereas for tapered circular ducts, the diameter of the section midway between large and small diameter shall be adopted, the length of tapered duct section shall be the centerline distance between the flanges of the duct section.

For special pieces like bends, tees, reducers, branches and collars, mode of measurement shall be identical to that described above using the length along the centerline.

The quoted unit rate for external surface of ducts shall include all wastage allowances, flanges and gaskets for joints, nuts and bolts, hangers and angles with double nuts for supports, rubber strip 5mm thick between duct and support, vibration isolator suspension where specified or required, inspection chamber/access panel, splitter damper with quadrant and lever for position indication, turning vanes, straightening vanes, and all other accessories required to complete the duct installation as per the specifications. These accessories shall NOT be separately measured nor paid for.

b. Special Items for Air Distribution shall be measured by the cross-section area perpendicular to air flow, as identified herewith :

i. Grilles and registers - width multiplied by height, excluding flanges. Volume control dampers shall form part of the unit rate for registers and shall not be separately accounted.

ii. Diffusers - cross section area for air flow at discharge area, excluding flanges. Volume control dampers shall form part of unit rate for supply air diffusers and shall not be separately accounted.

iii. Linear diffusers - shall be measured by cross-sectional areas and shall exclude flanges for mounting of linear diffusers. The supply air plenum for linear diffusers shall be measured with ducting as described earlier.

iv. Fire dampers - shall be measured by their cross sectional area perpendicular to the direction of air flow. Quoted rates shall include the necessary collars and flanges for mounting, inspection pieces with access door, electrical actuators and panel. No special allowance shall be payable for extension of cross section outside the air stream.

v. Flexible connection - shall be measured by their cross sectional area perpendicular to the direction of air flow. Quoted rates shall include the necessary mounting arrangement, flanges, nuts and bolts and treated-for-fire requisite length of canvas cloth.

vi. Kitchen Hoods - shall be measured by their cross sectional area at the capture point of fumes, parallel to the surface of kitchen equipment. Quoted rates shall include the grease filters, provision for hood light, suspension arrangement for the hood, profile to direct the air to ventilation ducts and provision for removable drip tray.

## 6.13 TESTING AND BALANCING

After the installation of the entire air distribution system is completed in all respects, all ducts shall be tested for air leaks by visual inspection. The entire air distribution system shall be balanced using an anemometer. Measured air quantities at fan discharge and at various outlets shall be identical to or less/excess than 5 percent of those specified and quoted. Branch duct adjustments shall be permanently marked after air balancing is completed so that these can be restored to their correct position if disturbed at any time. Complete air balance report shall be submitted for scrutiny and approval, and four copies of the approved balance report shall be provided with completion documents.

# 7.0 INSULATION

## 7.1 Scope

The scope of this section comprises the supply and application of insulation conforming to these specifications. The insulation material shall be Closed Cell Elastomeric Nitrile Rubber or Polyethylene Foam XLPE.

## 7.2 Material

Thermal insulation material for Duct insulation shall be with factory laminated black fiber glass cloth closed cell Nitrile rubber. Density of the nitrile rubber shall be 40-60 Kg/m$^3$
or
Al foil faced polyethylene material (XLPE). Density of the polyethylene material shall be 25-30 Kg/m$^3$

Comply

Thermal conductivity as per **BS 874 part 2 – 86 (DIN 52613, 52612) / DIN EN 12667 / EN ISO8497** of the insulation material shall not exceed 0.038 W/m$^o$K or 0.212 BTU / (Hr- ft$^2$-$^o$F/inch) at an average temperature of 30$^o$C. The product shall have temperature range of –40 $^o$C to 105$^o$C. The insulation material shall be fire rated for Class 0 as per BS 476 Part 6 : 1989 for fire propagation test and for Class 1 as per BS 476 Part 7, 1987 for surface spread of flame test. Water vapour permeability shall be not less than 0.024 per inch (2.48 x 10$^{-13}$ Kg/m.s.Pa i.e. $\mu \geq 7000$: Water vapour diffusion resistance) as per **DIN 53122 part 2, DIN 52615 / EN 12086 & EN13469**.

In addition to above properties the insulation material for ducts shall be anti-microbial. Microbiological growth on insulation surface shall be in accordance with ASTM G-21 and bacterial resistance to ASTM2180/ ISO22196.

The Material shall comply to ISO 5659 / BS 6853 / ABD 0031 for smoke density and toxicity values. The thermal conductivity of insulation material shall not be effected by aging as per **DIN 52616 standard.**

Thickness of the insulation shall be as specified for the individual application. **Each lot of insulation material delivered at site shall be accompanied with manufacturer's test certificate for density and thickness.** Adhesive used for sealing the insulation shall be non-flammable and with low VOC content (maximum 850 gm/l less water) as per IGBC guide lines) strictly as per manufacturer's recommendations. Adhesive shall be externally applied by contractor on site.

Ducting insulation thickness shall be as per table below.

Noted

| Ducting position | Thk. for non-coastal places |
|---|---|
| SA duct in RA path | 13 mm |
| Ducted return air system | SA duct: 19 mm<br>RA duct: 13 mm |
| Both SA & RA exposed | Both 25 mm |

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### 7.3 Duct Insulation

External thermal insulation shall be provided as follows :
The thickness of insulation material shall be as shown on drawings or identified in the schedule of quantity. Following procedure shall be adhered to:

Duct surfaces shall be cleaned to remove all grease, oil, dirt, etc. prior to carrying out insulation work.
Measurement of surface dimensions shall be taken properly to cut closed cell insulation to size with sufficient allowance in dimension. Cutting of insulation sheets shall be done with adjustable blade to make 90$^0$ cut in thickness of sheet. Hackshaw or blades are not acceptable tools for cutting the insulation.

Material shall be fitted under compression and no stretching of material shall be permitted. All longitudinal and transverse joints shall be sealed by providing 50 mm wide Fibre glass cloth laminated tape as per manufacturer recommendations. The insulation installers shall be certified by manufacture.

**Where ducts/pipes penetrates walls / floor it shall be insulated with intumescent properties insulation material for fire protection. The treatment shall be minimum 500 mm extended on both sides.**

### 7.4 Piping Insulation

**REFRIGERENT PIPING INSULATION**

Base insulation material for Refrigerant pipe shall be same as that of CHW pipe. The factory lamination on the base material shall be of chemically treated blue colored glass cloth of 7 mil thickness tested for UV resistance as per EN ISO 4892-2 Method-A Thickness of refrigerant piping insulation shall be 13mm and 19mm unless specified separately

**DRAIN PIPE INSULATION**

Noted

Base insulation material for drain pipe shall be same as that of CHW pipe. The factory lamination on the base material shall be of chemically treated blue colored glass cloth of 7 mil thickness tested for UV resistance as per EN ISO 4892-2 Method-A
Thickness of drain piping insulation shall be 13mm unless specified separately

### 7.5 Duct Acoustic Lining Open

**Nitrile Rubber**

Duct acoustic lining material shall be Nitrile Rubber open cell foam. Thermal conductivity of the insulation material shall not exceed 0.047 W/m$^o$K at an average temperature of 20$^o$C. Density of the nitrile rubber shall be 140 – 180 Kg/m$^3$. The material should withstand maximum surface temperature of +85$^o$C and minimum surface temperature of -20$^o$C. The material should conform to Class 1 rating for surface spread of Flame in accordance to BS 476 Part 7 & HBF, HF 1 & HF 2 in accordance to UL 94, 1996.

Comply

Insulation should have antimicrobial product protection, and should pass Fungi Resistance as per ASTM G 21 and Bacterial Resistance as per ASTM E 2180. The insulation should pass Air Erosion Resistance Test in accordance to ASTM Standard C 1071-05 (section 12.7).

Thickness of the material shall be 15 mm thick specified for the individual application and with noise absorption proprieties as per IS: 8225 / ISO 354 / ASTM423C. The insulation should be installed as per manufacturer's recommendation.

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### 7.6 Measurement Of Insulation

Unless otherwise specified measurement for duct and pipe insulation for the project shall be on the basis of center line measurements described herewith

7.6.1 Pipe Insulation shall be measured in units of length along the centre line of the installed pipe, strictly on the same basis as the piping measurements described earlier. The linear measurements shall be taken before the application of the insulation. It may be noted that for piping measurement, all valves, orifice plates and strainers are not separately measurable by their number and size. It is to be clearly understood that for the insulation measurements, all these accessories including cladding, valves, orifice plates and strainers shall be considered strictly by linear measurements along the centre line of pipes and no special rate shall be applicable for insulation of any accessories, fixtures or fittings whatsoever.

7.6.2 Duct Insulation and Acoustic Lining shall be measured on the basis of surface area along the centre line of insulation thickness. Thus the surface area of externally thermally insulated or acoustically lined be based on the perimeter comprising centre line (of thickness of insulation) width and depth of the cross section of insulated or lined duct, multiplied by the centre-line length including tapered pieces, bends, tees, branches, etc. as measured for bare ducting.

## 8.0 ELECTRICAL INSTALLATION

### 8.1 Scope

The scope of this section comprises of fabrication, supply, erection, testing and commissioning of Motor Control Centre (MCC), wiring and earthing of all air-conditioning equipment, components and accessories.

Note – Configuration of MCC panels shall be design to suit the requirement of system \ process. Necessary single line diagrams \ GA drawings shall be furnished by contractor for approval by consultant \ owner.

### 8.2 General

Work shall be carried out in accordance with the accompanying specifications and shall comply with the latest relevant Indian Standards and Electricity Rules and Regulations.

All motor control centres shall be suitable for operation on 3 Phase/single phase, 11,000/415/240 volts, 50 cycles, 4 wire system with neutral grounded at transformer. All MCCs be CPRI tested design and manufactured by a approved manufacturer. **CPRI certificate be made available.**

MCCs comply with the latest Relevant Indian Standards and Electricity Rules and Regulations and shall be as per IS-8623. MCCs / starter panels for outdoor equipment shall be suitable for outdoor duty application.

### 8.3 Constructional Features

The Motor Control Centre (MCC) shall be of 2 mm thick sheet steel cabinet and suitable for indoor installation, dead front, floor mounting/wall mounting type and shall be form 3b construction. The Distribution panels be totally enclosed, completely dust and vermin proof and be with hinged doors and folded covers, Neoprene gasket, padlocking arrangement and bolted back. All removable/ hinged doors and covers shall be groundedby flexible standard connectors. MCC shall be suitable for the climatic conditions as specified in Special Conditions. Steel sheets used in the construction of panels be 2 mm thick and be folded and braced as necessary to provide a rigid support for all components. Joints of any kind in sheet metal shall be seam welded, all welding, slag shall be rounded off and welding pits wiped smooth with plumber metal. The general construction confirm to IS-8623-1977 (Part-1) for factory built assembled switchgear & control gear for voltage upto and including 1100 V AC.

All MCCs/panels and covers shall be properly fitted and square with the frame, and holes in the panel correctly positioned. Fixing screws enter into holes tapped into an adequate thickness of metal or provided with wing nuts. Self threading screws not be used in the construction of Distribution panels. A base channel of 75 mm x 40 mm x 5 mm thick shall be provided at the bottom for floor mounted panels. Minimum **operating** clearance of 275 mm be provided between the floor of panels and the lowest operating height.

The MCCshall be of adequate size with a provision of spare feeders. Feeders bearranged in multi-tier. Knockout holes of appropriate size and number shall be providedin the Motor Control Centre in conformity with the location of cable/conduit connections. Removable sheet steel plates shall be provided at the top to make holes for additional cable entry at site if required.Every cabinet shall be provided with Trifoliate or engraved metal name plates. All panels shall be provided with circuit diagram mounted on inside of door shutter protected with Hylam sheet. All live accessible connections shall be shrouded and minimum clearance between phase and earth be 20 mm and phase to phase be 25 mm.

Panels with ACB shall necessarily have front and rear access as per requirement whereas panels with all MCCB breaker shall be provided with front access with sufficient clearance

**8.4 Wiring System**

All control wiring shall be carried out by using PVC insulated copper conductor wires in conduits. The minimum size of control wiring be 1.5 sq mm. Minimum size of conductor for power wiring shall be 4 sq. mm 1100 volts grade PVC insulated copper conductor wires in conduit. All conductors shall be stranded.

**8.5 Circuit Compartment**

All components for each feeder shall be housed in a separate compartment and have steel sheets on top and bottom of compartment. Sheet steel hinged lockable door beduly interlocked with the breaker in the “ON” position. Safety interlocks be provided to prevent the breaker from being drawn-out when the breaker is in ‘ON’ position. The door not form an integral part of the draw-out portion of the panel. Sheet steel barriers shall be provided between the tiers in a vertical section.

All MCCs shall be provided with feeders of appropriate capacity as per Single Line Diagram. All MCCs shall be completely factory wired, ready for connection. All the terminals shall be of proper current rating and sized to suit individual feeder requirements. Each circuit be clearly numbered from left to right to correspond with wiring diagram. All the switches and circuits be distinctly marked with a small description of the service installed.

Continuous earth bus sized for prospective fault current shall be provided with arrangement for connecting to station earth at two points. Hinged doors/ frames shall be connected to earth through adequately sized flexible braids.

### 8.6 Instrument Accommodation

Adequate space shall be provided for accommodating instruments, indicating lamps, control contactors and control MCBs. These shall be accessible for testing and maintenance without any danger of accidental contact with live parts of the circuit breaker and bus bar ‘ON’ lamps shall be provided on all outgoing feeders.

### 8.7 Bus Bar Connections

Bus bar and interconnections shall be of high conductivity electrolytic grade aluminium/copper complying with requirement of IS : 5082 – 1981 and of rectangular cross section suitable for carrying the rated full load current and short circuit current and shall be extendable on either side. Copper conductor shall be used for busbar of rating 1000A and above. Bus bars and interconnections shall be insulated with heat shrinkable sleeve of 1.1 KV grade and shall be colour coded. Bus bars shall be supported on glass fiber reinforced thermosetting plastic insulated supports at regular intervals to withstand the force arising from in case of short circuit in the system. All bus bars shall be provided in a separate chamber and all connections shall be done by bolting. Additional cross sectional area to be added to the bus bar to compensate for the holes. All connections between bus bars and breakers shall be through solid copper / aluminium strips of proper size to carry full rated current and insulated with insulating sleeves. Maximum current density for the busbars be 0.8 A/sq.mm for aluminium and 1.4A/sq.mm for copper busbars.

**Maximum allowable temperature for the Bus bar to be restricted to 85 deg C**

### 8.8 Temperature – Rise Limit

Unless otherwise specified, in the case of external surface of enclosures of bus bar compartment which shall be accessible but do not need to be touched during normal operation, an increase in the temperature rise limits of 25° C above ambient temperature be permissible for metal surface and of 15° C above ambient temperature for insulating surfaces as per IS 8623(Part-2) 1993.

### 8.9 Cable Compartments

Cable compartment of adequate size shall be provided in the panel for easy clamping of all incoming and outgoing cables entering from the top/bottom. Adequate supports be provided in cable compartment to support cables as per approved for construction shop drawing.

## 8.10 AIR CIRCUIT BREAKERS (ACB)

The ACB conform to the requirements of IEC 60947-2 / IS 13947-2 and shall be type tested & certified for compliance to standards from–CPRI, ERDA/ any accredited international lab. The circuit breaker shall be suitable for 415 V $\pm$ 10%, 50 Hz supply system. Air Circuit Breakers be with moulded housing flush front, draw out type and shall be provided with a trip free manual operating mechanism or as indicated in drawings with mechanical "ON" "OFF" “TRIP” indications.

The ACB be 3/ 4 pole with modular construction, draw out, manually or electrically operated version as specified. The circuit breakers shall be for continuous rating and service short Circuit Breaking capacity (Ics) shall be as specified on the single line diagram and should be equal to the Ultimate breaking capacity(Icu) and short circuit withstand values(Icw) for 1 sec.

Circuit breakers shall be designed to ‘close’ and ‘ trip’ without opening the circuit breaker compartment door. The operating handle and the mechanical trip push button shall be at the front of the breakers panel. Inspection of main contacts should be possible without using any tools. The ACB shall be provided with a door interlock. i.e. door should not be open when circuit breaker is closed and breaker should not be closedwhen door is open.

All current carrying parts shall be silver plated and suitable arcing contacts with proper arc chutes shall be provided to protect the main contacts. The ACB have double insulation (Class-II) with moving and fixed contacts totally enclosed for enhanced safety and in accessibility to live parts. All electrical closing breaker be with electrical motor wound stored energy spring closing mechanism with mechanical indicator to provide ON/OFF status of the ACB.

The auxiliary contacts blocks shall be so located as to be accessible from the front. The auxiliary contacts in the trip circuits close before the main contacts have closed. All other contacts close simultaneously with the main contacts. The auxiliary contacts in the trip circuits open after the main contacts open. Minimum 4 NO and 4 NC auxiliary contacts be provided on each breaker.

Rated insulation voltage be 1000 volts AC.

### 8.10.1 Cradle

The cradle shall be so designed and constructed as to permit smooth withdrawal and insertion of the breaker into it. The movements be free from jerks, easy to operate and be on steel balls/rollers and not on flat surfaces.

There shall be 4 distinct and separate position of the circuit breaker on the cradle. Racking Interlock in Connected/Test/Disconnected Position.

Service Position : Main Isolating contacts and control contacts of the breaker are engaged.

Test Position : Main Isolating contacts are isolated but control contacts are still engaged.

Isolated Position : Both main isolating and control contacts are isolated.

There shall be provision for locking the breaker in any or all of the first three positions.

The following safety features be incorporated :

8.10.1.1 Withdrawal or engagement of Circuit breaker not be possible unless it is in opencondition.

8.10.1.2 Operation of Circuit breaker not be possible unless it is fully in service, test ordrawn out position.

8.10.1.3 All modules shall be provided with safety shutters operated automatically by movement of the carriage to cover exposed live parts when the module iswithdrawn.

8.10.1.4 All Switchgear module front covers have provision for locking.

8.10.1.5 Switchgear operating handles shall be provided with arrangement for locking in‘OFF’ position.

8.10.2 Protections

The breaker should be equipped with micro-controller based , communicable type release with RS 485 port for communication to offer accurate and versatile protection withcomplete flexibility and offer complete over current protection to the electrical system in the following four zones :

- Long time protection.
- Short time protection with intentional delay.
- Instantaneous protection.
- Ground fault protection.

The protection release generally have following features and settings **<u>however for exact selection of protection releases shall be made based on project requirement.</u>**

8.10.2.1 True RMS Sensing
The release sample the current at the rate of 16 times per cycle to monitor the actual load current waveform flowing in the system and monitor the true RMS value of the load current. It take into account the effect of harmonics also.

8.10.2.2 Thermal Memory
When the breaker reclose after tripping on overload, then the thermal stresses caused by the overload if not dissipated completely, get stored in the memory of the release and this thermal memory ensure reduced tripping time in case of subsequent overloads. Realistic Hot/Cold curves take into account the integrated heating effects to offer closer protection to the system.
8.10.2.3 Defined time-current characteristics :

A variety of pick-up and time delay settings shall be available to define the current thresholds and the delays to be set independently for different protection zones thereby achieving a close-to-ideal protection curve.

8.10.2.4Trip Indication

Individual fault indication for each type of fault should be provided by LEDs for faster fault diagnosis.

8.10.2.5Self powered

The release draw its power from the main breaker CTs and require no external power supply for its operation.

8.10.2.6Zone Selective Interlocking

The release shall be suitable for communication between breakers to enable zone selective interlocking. This feature shall be provided for both short circuit and ground fault protection zones to offer intelligent discrimination betweenbreakers. This feature enables faster clearance of fault conditions, thereby reducing the thermal and dynamic stresses produced during fault conditions and thus minimises the damage to the system.To implement ZSI manufacturer shouldsupply all related equipment like power supply, wiring etc.

On-Line change of settings should be possible. It should be possible to carry outtesting of release without tripping the breaker.

8.10.2.7The release meet the EMI / EMC requirements.

8.10.2.8The setting range of release shall be generally as follows :

| | SETTING RANGE OF RELEASE | |
|---|---|---|
| Type of Protection | PICK-UP CURRENT | TIME DELAY |
| Long Time | 0.4 to 1.0 times $I_n$ ($I_r$)<br><br>Steps : 0.4, 0.5, 0.55, 0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90, 0.95, 1.00.<br><br>Operating Limit : 1.05 to 1.2times $I_r$ | 0.5 to 30 sec at 6 $I_r$<br><br>Steps 0.5,1, 2,4, 6, 8,12,18,24 and 30 secs<br><br>Tolerance : Corresponding to ±10% of current. |
| Short Time | 2 to 10 times $I_r$<br><br>Steps : 2,3,4,5,6,7,8,9 & 10<br><br>Tolerance : ±10% | 20 ms to 600 ms<br>Steps<br>20,60,100,160,200,260,300 400,500 and 600 ms Tolerance : ±10% or 20ms<br>whichever is higher |
| Instantaneous | 2 to 12 times $I_n$<br>Steps : 2,3,4,6,8,10,12<br>Tolerance : ±10% | |
| Ground Fault | 0.2 to 0.6 time $I_n$<br><br>Steps : 0.2,0.3,0.4,0.5,0.6<br><br>Tolerance : ±10% | 100 ms to 400 ms<br><br>Steps : 100,200,300,400ms<br>Tolerance : ±10% or 20 ms whichever is higher. |

All **incomer** ACBs have following additional protections other than mentioned above.

8.10.2.8.1 Under and over voltage

8.10.2.8.2 Under and over frequency

8.10.2.8.3 Restricted Earth Fault protection

8.10.2.8.4 Trip Circuit supervision with PS class CT's.

8.10.2.8.5 Undercurrent, ( for DG set only)

8.10.2.8.6 Reverse power ( for DG set only)

8.10.2.8.7 Phase sequence reversal ( for DG set only)

8.10.2.8.8 Load shedding and reconnection thru programmable contacts.

8.10.2.8.9 Release should display the Contact wear indication.

The release should provide local indication of actual %age loading at any instant. The release should be able to communicate on MODBUS RTU protocol using inbuilt RS485/232/Ethernet port and shall be integral part of supply with trip unit. Parameters of the Protection Release should be changeable from Release as well as thrucommunication network. Release should have graphical LCD for display of power parameters. The release of incoming breakers should provide comprehensive metering with the following parameters

8.10.2.8.10 Phase currents (running, avg & max) – All parameters in single window.

8.10.2.8.11 Release should be able to capture short circuit current on which ACB has tripped. The last ten trips and alarms shall be stored in memory with the date & time stamping along with type of fault and alarm. The sensing CT Should be Rogowsky type with measurement precision of 1%.

8.10.2.8.12 Release should be self powered .

8.10.2.8.13 Release should have facility to select different type of IDMTL protection(DT,SIT,VIT,EIT,HVF) for better co-ordination with HT Breaker/Fuse.

8.10.2.8.14 Phase voltages (running, avg & max)

8.10.2.8.15 Energy & power parameters (active, reactive and apparent)

8.10.2.8.16 PF

8.10.2.8.17 Frequency

8.10.2.8.18 Maximum Demand ( KVA & KW)

8.10.2.8.19 Total Harmonics

distortion All **O/G** ACBs have following

functions.

**Protection**

- The ACB control unit offer the following protection functions as standard: Long-time (LT) protection with an adjustable current setting and time delay;
- Short-time (ST) protection with an adjustable pick-up and time delay;
- instantaneous (INST) protection with an adjustable pick-up and an OFF Position.
- Current and time delay setting be indicated in amperes and seconds respectively On a digital display.
- Earth-fault protection with an adjustable pick-up and time delay be provided if indicated on the appended single-line diagram.

**Measurements**

- An ammeter with a digital display indicate the true rms values of the currents for each phase. Release acknowledge the current & time delay settings done by user on the LCD display.
- A LED bargraph simultaneously display the load level on the three phases.
- A maximeter store in memory and display the maximum current value observed since the last reset. The data continue to be stored and displayed even after openingof the circuit breaker.

8.10.3 Safety Features

I. The safety shutter prevent inadvertent contact with isolating contacts when breaker is withdrawn from the Cradle.

II. It not be possible to interchange two circuit breakers of two different thermal ratings. For Draw-out breakers, an arrangement be provided to prevent rating mismatch between breaker and cradle.

III. There shall be provision of positive earth connection between fixed and moving portion of the ACB either thru connector plug or sliding solid earth mechanism. Earthing bolts shall be provided on the cradle or body of fixed ACB.

IV. The incoming panel accommodating ACB shall be provided with indicating lamps for ON-OFF positions, digital voltmeter and ammeter of size not less than 96 mm x 96 mm, selector switches, MCB for protection circuit and measuring instrument circuits.

V. It shall be possible to bolt the draw out frame not only in connected position but also in TEST and DISCONNECTED position to prevent dislocation due to vibration and shocks.

VI. Draw out breakers should not close unless in distinct Service/Test/Isolated positions.

VII. The insulation material used conform to Glow wire test as per IEC60695.

VIII. The ACB provide in built electrical and mechanical anti-pumping.

IX. All EDO ACB‘ s have Ready to Close Contact to ensure that the ACB gets a command only when it is ready to close for applications of Remote Control, AMF, Synchronization and Auto Source Change Over Systems.

## 8.11 MOULDED CASE CIRCUIT BREAKER (MCCB)

The MCCB should be current limiting type with trip time of less than 10 msec under short circuit conditions. The MCCB should be either 3 or 4 poles. MCCB comply with the requirements of the relevant standards IS13947 – Part 2/IEC 60947-2 and should have test certificates for Breaking capacities from independent test authorities CPRI / ERDA or any accredited international lab.

MCCB comprise of Quick Make -break switching mechanism, arc extinguishing device and the tripping unit shall be contained in a compact, high strength, heat resistant, flame retardant, insulating moulded case with high withstand capability against thermal and mechanical stresses

The breaking capacity of MCCB be as specified in the Drawings. The rated service breaking capacity (Ics) should be equal to rated ultimate breaking capacities (Icu).MCCBs for motor application should be selected in line with Type-2 Co-ordination as per IEC-60947-2, 1989/IS 13947-2. The breaker as supplied to meet IP54 degree of protection.

### 8.11.1 Current Limiting & Coordination

- The MCCB employ maintenance free minimum let-through energies and capable of achieving discrimination up to the full short circuit capacity of the downstream MCCB. **The manufacturer provide both the discrimination tables and let-through energy curves for all.**

  **Protection Functions**

  - MCCBs with ratings less than 100 A shall be equipped with Thermal-magnetic (**adjustable** thermal for overload and **fixed** magnetic for short-circuit protection) trip units
  - Microprocessor MCCBs with ratings 100A and above shall be equipped with microprocessor based trip units.
  - Microprocessor and thermal-magnetic trip units shall be adjustable and it shall be possible to fit lead seals to prevent unauthorized access to the settings
  - Microprocessor trip units comply with appendix F of IEC 60947-2 standard (measurement of RMS current values, electromagnetic compatibility, etc.)
  - Protection settings apply to all poles of circuit breaker.
  - All Microprocessor components withstand temperatures up to 125 °C

### 8.11.2 Testing

a) Original test certificate of the MCCB as per IEC 60947-1 &2 or IS13947 be furnished.

b) Pre-commissioning tests on the switch board panel incorporating the MCCB shall be done as per standard specifications.

### 8.11.3 Interlocking

Moulded, case circuit breakers be provided with the following interlocking devices for interlocking the door of a switch board.

a) Handle interlock to prevent unnecessary manipulations of the breaker.

b)Door interlock to prevent the door being opened when the breaker is in ON position.

c) Defeat-interlocking device to open the door even if the breaker is in ON position.

- The MCCB shall be current limiting type and comprise of quick make – Break switching mechanism. MCCBs shall be capable of defined variable overload adjustment. All MCCBs rated 100 Amps and above have adjustable over load & short circuit pick-up .

  - All MCCB with microprocessor based release unit, the protection be adjustable Overload, Short circuit and earth fault protection with time delay.

  The trip command override all other commands.

## 8.12 MOTOR PROTECTION CIRCUIT BREAKER (MPCB)

Motor circuit breakers conform to the general recommendations of standard IEC 947 -1,2 and 4 (VDE 660, 0113 NF EN 60 947-1-2-4, BS 4752) and to standards UL 508 and CSA C22-2 N°14.

The devices shall be in utilization category A, conforming to IEC 947-2 and AC3 conforming to IEC 947-4.MPCB have a rated operational and insulation voltage of 690V AC (50 Hz) and MPCB shall be suitable for isolation conforming to standard IEC 60947-2 and have a rated impulse withstand voltage (Uimp) of 6 kV. The motor circuit breakers

shall be designed to be mounted vertically or horizontally without de-rating. Power supply be from the top or from the bottom. In order to ensure maximum safety, the contacts shall be isolated from other functions such as the operating mechanism, casing, releases, auxiliaries, etc, by high performance thermoplastic chambers. The operating mechanism of the motor circuit breakers must have snap action opening and closing withfree tripping of the control devices. All the poles close, open, and trip simultaneously.
The motor circuit breakers accept a padlocking device in the “isolated” position.

The motor circuit breakers shall be equipped with a “PUSH TO TRIP” device on the front enabling the correct operation of the mechanism and poles opening to be checked. The auxiliary contacts shall be front or side mounting, and both arrangements be possible. The front-mounting attachments not change the breaker surface area. Depending on its mounting direction the single pole contact block could be NO or NC. All the electrical auxiliaries and accessories shall be equipped with terminal blocks and shall be plug-in type. The motor circuit breakers have a combination with the downstream contactor enabling the provision of a perfectly co-ordinated motor-starter. This combination enable type 1 or type 2 co-ordination of the protective devices conforming to IEC 60947-4- 1.Type 2 co-ordination be guaranteed by tables tested and certified by an official laboratory: LOVAG (or other official laboratory).The motor circuit breakers, depending onthe type, could be equipped with a door-mounted operator which allow the devicesetting. The motor circuit breakers shall be equipped with releases comprising a thermal element assuring overload protection and a magnetic element for short-circuit protection.In order to ensure safety and avoid unwanted tripping, the magnetic trip threshold (fixed) be factory set to an average value of 12 Ir.

All the elements of the motor circuit breakers shall be designated to enable operation at an ambient temperature of 60°C without derating. The thermal trips shall be adjustable on the front by a rotary selector. The adjustment of the protection shall be simultaneous for all poles. Phase unbalance and phase loss detection shall be available. Temperature compensation (-20°C to +60°C)

### 8.13 MINIATURE CIRCUIT BREAKER (MCB)

Miniature Circuit Breaker comply with IS-8828-1996/IEC898-1995. Miniature circuit breakers shall be quick make and break type for 240/415 VAC 50 Hz application with magnetic thermal release for over current and short circuit protection. The breaking capacity not be less than 10 KA at 415 VAC. MCBs shall be DIN mounted. The MCB shall be Current Limiting type (Class-3). MCBs shall be classified (B,C,D ref IS standard) as per their Tripping Characteristic curves defined by the manufacturer. The MCB have the minimum power loss (Watts) per pole defined as per the IS/IEC and the manufacturer publish the values.MCB ensure complete electrical isolation & downstream circuit or equipment when the MCB is switched OFF.

The housing shall be heat resistant and having a high impact strength. The terminals shall be protected against finger contact to IP20 Degree of protection. All DP, TP, TPN and 4 Pole miniature circuit breakers have a common trip bar independent to theexternal operating handle.

### 8.14 Painting

All sheet steel work undergo a process of degreasing, pickling in acid, cold rinsing, phosphating, passivaiting (seven tank processing) and then painted with electrostatic paint (Powder coating). The shade of colour of panel inside/outside shall be as indicated in datasheets & relevant BIS code.

### 8.15 Labels

Engraved PVC labels shall be provided on all incoming and outgoing feeder. Circuit diagram showing the arrangements of the circuit inside the control panel shall be pasted on inside of the panel door and covered with transparent plastic sheet.

### 8.16 Meters

i. All voltmeters and indicating lamps shall be through MCB's.

ii. Meters and indicating instruments be plug type.

iii. All CT's connection for meters shall be through Test Terminal Block (TTB).

iv. CT ratio and burdens shall be as specified on the Single line diagram.

### 8.17 Current Transformers

Current transformers are provided for Control panels carrying current in excess of 60 amps. All phase be provided with current transformers of suitable VA burden with5 amps secondaries for operation of associated metering.

The CTs confirm to relevant Indian Standards. The design and construction shall be dry type, epoxy resin cast**,** robust to withstand thermal and dynamic stresses during short circuits. Metering CTs, have inbuilt busbar mounting arrangement. Secondary terminals of CTs be brought out suitable to a terminal block which be easily accessible for testing and terminal connections. The secondary terminal should be covered with insulation cap/cover so that there should not be any possibility of touching the live terminal. The protection CTs be of accuracy class 5P20 and measurement CTs be of accuracy classI.

### 8.18 Selector Switch

Where called for, selector switches of rated capacity be provided in control panels, to give the choice of operating equipment in selective mode.

### 8.19 Contractor

Contactor shall be built into a high strength thermoplastic body and shall be provided with an arc shield for quick arc extinguishing. Silver alloy tips shall be provided to ensure a high degree of reliability and endurance under continuous operation. The magnet system consist of laminated yoke and armature to ensure clean operation without hum orchatter.

Starters contactors have 3 main and 2 Nos. NO / NC auxiliary contacts and shall be air break type suitable for making and breaking contact at minimum power factor of 0.35. For design consideration of contactors the starting current of connected motor shall be assumed to be 6 times the full load current of the motor in case of direct-on-line starters and 3 times the full load current of the motor in case of Star Delta and Reduced Voltage Starters. The insulation for contactor coils be of Class “E”.

Coil shall be tape wound vacuum impregnated and be housed in a thermostatic bobbin, suitable for tropical conditions and withstand voltage fluctuations. Coil be suitable for 220/415±10% volts AC, 50 cycles AC supply.

### 8.20 Thermal Overload Relay

Thermal over load relay have built in phase failure sensitive tripping mechanism to prevent against single phasing as well as on overloading. The relay operate on the differential system of protection to safeguard against three phase overload, single phasing and unbalanced voltage conditions.

Auto-manual conversion facility shall be provided to convert from auto-reset mode to manual-reset mode and vice-versa at site. Ambient temperature compensation shall be provided for variation in ambient temperature from -5° C to +55°C.

All overload relays shall be of three elements, positive acting ambient temperature compensated time lagged thermal over load relays with adjustable setting. Relays shall be directly connected for motors up to 35 HP capacity. C.T. operated relays be provided for motors above 35 HP capacity. Heater circuit contactors may not be provided with overload relays.

### 8.21 Time Delay Relays

Time delay relays shall be adjustable type with time delay adjustment from 0-180 seconds and have one set of auxiliary contacts for indicating lamp connection.

### 8.22 Indicating Lamp And Metering

All meters and indicating lamps be in accordance with relevant IS standard specification. The meters shall be flush mounted type. The indicating lamp shall beof LED type . Each MCC and control panel be provided with voltmeter 0-500 volts with three way and off selector switch, CT operated ammeter of suitable range with threenos. CTS of suitable ratio with three way and off selector switch, phase indicating lamps, and other indicating lamps as called for. All indicating lamp be backed up with 5 amps MCB.

### 8.23 Toggle Switch

Toggle switches, where required, shall be in conformity with relevant IS Codes and be of 5 amps rating.

### 8.24 Push Button Stations

Push button stations shall be provided for manual starting and stopping of motors / equipment Green and Red colour push buttons shall be provided for 'Starting' and 'Stopping' operations. 'Start' or 'Stop' indicating flaps shall be provided for push buttons. Push Buttons shall be suitable for panel mounting and accessible from front without opening door, Lock lever be provided for 'Stop' push buttons. The push button contacts be suitable for 6 amps current capacity.

### 8.25 Conduits

Conduits and Accessories conform to latest edition of Indian Standards IS-9537 part 1 & 2. 16/14 (16 gauge upto 32mm diameter & 14 gauge above 32 mm diameter) gauge screwed GI or MS conduits to be used. Joints between conduits and accessories shall be securely made by standard accessories, as per IS-2667, IS-3837 and IS-5133 to ensure earth continuity. All conduit accessories shall be threaded type only.

Only approved make of conduits and accessories be used.

Conduits shall be delivered to the site of construction in original bundles and each length of conduit bear the label of the manufacturer.

**Note.:** Whatever materials required to be billed by the Contractor should come on site with proper Challan Numbers and quantity mentioned in each such Challan.

Maximum permissible number of 1100 volt grade PVC insulated wires that may be drawn into metallic Conduits are given below :

| Size of wires Nominal Cross section Area (Sq. mm.) | Maximum number of wires within conduit size(mm) | | | | |
|---|---|---|---|---|---|
| | 20 | 25 | 32 | 40 | 50 |
| 1.5 | 5 | 10 | 14 | -- | -- |
| 2.5 | 5 | 8 | 12 | -- | -- |
| 4 | 3 | 7 | 10 | -- | -- |
| 6 | 2 | 5 | 8 | -- | -- |
| 10 | -- | 3 | 5 | 6 | -- |
| 16 | -- | 2 | 3 | 6 | 6 |
| 25 | -- | -- | 2 | 4 | 6 |
| 35 | -- | -- | -- | 3 | 5 |

Maximum permissible number of 1100 volt grade PVC insulated wires that may be drawn into rigid non metallic or PVC Conduits are given below :

| Size of wires Nominal Cross section Area (Sq. mm.) | Maximum number of wires within conduit size(mm) | | | | |
|---|---|---|---|---|---|
| | 20 | 25 | 32 | 40 | 50 |
| 1.5 | 7 | 12 | 16 | -- | -- |
| 2.5 | 5 | 10 | 14 | -- | -- |
| 4 | 4 | 8 | 12 | -- | -- |
| 6 | 3 | 6 | 8 | -- | -- |
| 10 | -- | 4 | 5 | 6 | -- |
| 16 | -- | 3 | 3 | 6 | 6 |
| 25 | -- | -- | 2 | 4 | 6 |
| 35 | -- | -- | -- | 3 | 5 |

**8.26 Cables**

1100V grade Cables of sizes 25 sq. mm. and above shall be XLPE FRLS insulated aluminium conductor armoured type and PVC insulated Copper conductor armoured cables for sizes 16 sq. mm. and below. All cables shall be conforming to IS Codes. Cables shall be suitable for laying in trenches, ducts, and on cable trays as required. Cables shall be termite resistant. Cable glands shall be heavy duty double compression brass glands. Control cables and indicating panel cables shall be multi core PVC insulated copper conductor and armoured cables.

The equipment inside plant room shall be connected to the control panel by means of suitable cables of adequate size. An isolator shall be provided near each motor/equipment (mounted within 10 ~ 15 mtr distance on nearest wall or self supported on floor ) wherever the motor/equipment is separated from the supply panel through a partition barrier or through ceiling construction. PVC insulated copper conductor wires shall be used inside the control panel for connecting different components and all the wires inside the control panel shall be neatly dressed and plastic beads shall be providedat both the ends for easy identification of control wiring.

Cables shall be cross linked polyethylene (XLPE) insulated PVC inner sheathed and FRLS PVC outer sheath of 1100 volts grade. Cables shall be suitable for laying in trenches, ducts, and on cable trays as required. Cables shall be termite resistant. Cable glands shall be double compression glands. Control cables and indicating panel cables shall be multi core PVC insulated copper conductor and armoured cables. All conductors shall be stranded.

Cabling for following equipment shall be fire survival type.

1. Basement smoke exhaust fan and make up air
2. Smoke evacuation fan
3. Staircase, lift, lobby pressurization fan
4. Make-up air fan for emergency duty

### 8.27 Fire Survival Armoured Cable

Fire Survival armoured cable, LPCB / BRE-GLOBAL / ERDA approved, with class -2, annealed copper / or aluminum conductor having glass mica fire barrier tape extruded with cross linkable low smoke zero halogen insulation. The inner & outer sheath shall be LSZH. The basic design shall be as per BS: 7846 & BS: 5839-1 (Latest edition).
The cable should meet fire performance circuit integrity test as per BS 8434-2 / BS 6387 CWZ.

### 8.28 Fire Survival Un-armoured Cable

Fire Survival Un-armoured cable, LPCB / BRE-GLOBAL / ERDA approved, with class -2, annealed copper or aluminum conductor having glass mica fire barrier tape extruded with cross linkable low smoke zero halogen insulation. The outer sheath shall be LSZH. The basic design shall be as per BS: 7629 & BS: 5839-1 (Latest edition).
The cable should meet fire performance circuit integrity test as per BS 8434-2 / BS 6387 CWZ.
Cabling shall be of the following sizes as minimum :

| | | |
|---|---|---|
| i. | From 30 HP to 35 HP motors | 2 nos. 3 x 16 sq. mm aluminium conductor armoured cable. |
| ii. | From 40 HP to 50 HP motors | 2 Nos. 3 x 25 sq. mm. aluminium conductor armoured cable. |
| iii. | From 60 HP to 75 HP motors | 1 No. 3 x 70 sq. mm aluminium conductor armoured cable. |
| iv. | 100 HP motors | 1 No. 3 x 150 sq. mm. aluminium conductor armoured cable |
| v. | 150 HP motor | 1 No. 3 x 240 sq. mm. aluminium conductor armoured cable. |
| vi. | 250 HP motor | 2 Nos. 3 x 240 sq. mm. aluminium conductor armoured cable. |
| vii. | 400 HP motor | 3 Nos. 3 x 240 sq. mm. aluminium conductor armoured cable |

| | | |
|---|---|---|
| viii. | 600 HP motor | 3 Nos. 3 x 400 sq. mm. aluminium conductor armoured cable. |

**HVAC contractor shall submit cable schedule for all equipment for approval.**

**8.29 Cable Laying**

Cables shall be laid by skilled and experienced workmen using adequate rollers to minimize stretching of the cable. The cable drums be placed on jacks beforeunwinding the cable. Great care shall be exercised in laying cables to avoid forming kinks.

8.29.1 Laying of Cables on Cable Trays

The relative position of the cables, laid on the cable tray be preserved and the cables not cross each other. At all changes in direction in horizontal and vertical planes, the cable shall be bent smooth with a radius as recommended by the manufacturer's. All cables be laid with minimum one diameter gap and shall be clamped at every metre to the cable tray. Cables be tagged for identification with aluminum tag and clamped properly at every 20M. Tags be provided at both ends and all changes in directions both sides of wall and floor crossings. All cable be identified by embossing on the tag the size of the cable, place of origin and termination.

All cables passing through holes in floor or walls shall be sealed with fire retardant Sealant and shall be painted with fire retardant paint upto one meter on all joints,terminations and both sides of the wall crossings by "VIPER CABLE RETARD".

8.29.2 Laying of Cables in Ground

The width of trench for laying single cable be minimum 350 mm. Where more than one cable is to be laid in horizontal formation, the width of the trench be workout by providing 200 mm gap between the cables, except where otherwise specified. There shall be clearance of 150 mm between the end cable and the side wall of the trench. Theminimum depth of the cable trench no to be less than 750 mm for single layer of cables. When the cables are laid in more than one tier the depth of the trench shall be increased by 300 mm for each additional tier.

<u>Excavation of trenches</u> : The trenches shall be excavated in reasonably straight lines. Wherever there is a change in direction, suitable curvature shall be provided. Where gradients and changes in depth are unavoidable, these shall be gradual. The excavated soil shall be stacked firmly by the side of the trench such that it may not fall back into the trench. The bottom of the trench shall be levelled and shall be made free from stone, brick bats etc. The trench then shall be provided with a layer of clean, dry sand cushion of not less than 100 mm in depth. Prior to laying of cables, the cores shall be tested for continuity and insulation resistance. The cable drum be properly mounted on jacks, at a suitable location, making sure that the spindle, jack etc. are strong enough to carry the weight of the drum and the spindle is horizontal. Cable shall be pulled over rollers in the trench steadily and uniformly without jerks and strains. The entire drum length shall be laid in one stretch. However, where this is not possible the remainder of the cable be removed by 'Flaking' i.e. by making one longloop in the reverse direction. After the cable has been uncoiled and laid into the trench over the rollers, the cable shall be lifted off the rollers beginning from one endby helpers standing about 10 meters apart and laid in a reasonably straight line. Cable laid in trenches in a single tier formation have a cover of clean, dry sand of not less than 150 mm. above the base cushion of sand before the protective cover is laid. In the case of vertical multi-tier formation after the first cable has been laid, a sand cushion

of 300 mm shall be provided over the initial bed before the second tier is laid. Finally the cables be protected by second class bricks before back filling the trench. The buried depth of uppermost layer of cable not be less than 750mm.

Back Filling : The trenches shall be back filled with excavated earth free from stones or other sharp edged debris and shall be rammed and watered, if necessary, in successive layers not exceeding 300 mm. Unless otherwise specified, a crown of earth not less than 50 mm in the centre and tapering towards the sides of the trench shall be left to allow for subsidence.

8.29.3 WIRE AND WIRE SIZES

1100 volts grade PVC insulted copper conductor wires in conduit shall be used.

For all single phase/ 3 phase wiring, 1100 volts grade PVC insulated copper conductor LSZH wires shall be used. The equipment inside plant room and AHU room shallbe connected to the control panel by means of insulated copper conductor wires of adequate size in exposed conduits. Final connections to the equipment shall be through wiring enclosed in galvanized flexible conduits rigidly clamped at both ends and at regular intervals. An isolator shall be provided near each motor/equipment wherever the motor/equipment is separated from the supply panel through a partition barrier or through ceiling construction. PVC insulated copper conductor wires shall be used inside the control panel for connecting different components and all the wires inside the control panel shall be neatly dressed and plastic beads be provided at both the ends for easy identification of control wiring.

The minimum size of control wiring shall be 1.5 sq. mm PVC insulated stranded soft drawn copper conductor wires drawn through conduit to be provided for connecting equipment and control panels.

Power cabling shall be of the minimum following sizes:

| | | |
|---|---|---|
| i. | Upto 5 HP motors/ 5 KW heaters | 3C x 4 sq. mm copper conductor PVC insulated cables. |
| ii. | From 6 HP to 10 HP motors<br>6 KW to 7.5 KW heaters | 3 x 6 sq. mm copper<br>conductor PVC insulated cables. |
| iii. | From 12.5 HP to 15 HP<br>motors | 2 Nos. 3 x 6 sq. mm<br>copper conductor PVC insulated cables. |
| iv. | From 20 HP to 25 HP motors | 2 Nos. 3 x 10 sq. mm copper conductor PVC insulated cables |

**Starters**

Each motor shall be provided with a starter of suitable rating. Starters be in accordance with relevant IS Codes. All Star Delta Starters be fully automatic. **Motors up to 7.5 HP be provided by Direct On Line (DOL) starter, motors above 7.5 HP andup to 45 HP shall be provided by star/delta starter and motors above 45 HP shall beprovided by soft starter**. All starters be with Type II coordination for breaker, contactor and over load relay.

All the switches, contactors, push button stations, indicating lamps be distinctly marked with a small description of the service installed. The following capacity contactors and overload relays shall be provided for different capacity motors or as per manufacturer's recommendation.

--------------------------------------------------------------------------------------------------------

| TYPE OF STARTER CAPACITY | | CONTACTOR CURRENT RANGE | OVERLOADRELAY | |
|---|---|---|---|---|
| 5 | HP Motors | D O L | 16 amps | 6-10 amps |
| 7.5 | HP motors | D O L | 16 amps | 9-15 amps |
| 10 | HP Motors | Automatic Star Delta | 25 amps | 9-15 amps |
| 12.5 | HP Motors | Automatic Star Delta | 16 amps | 9-15 amps |
| 15 | HP Motors | Automatic Star Delta | 25 amps | 9-15 amps |
| 20 | HP Motors | Automatic Star Delta | 32 amps | 14-23 amps |
| 25 | HP Motors | Automatic Star Delta | 32 amps | 14-23 amps |
| 30 | HP Motors | Automatic Star Delta | 40 amps | 20-33 amps |
| 35 | HP Motors | Automatic Star Delta | 40 amps | 20-33 amps |
| 40 | HP Motors | Automatic Star Delta | 40 amps | 30-50 amps |
| 50 | HP Motors | VFD | 70 amps | 30-50 amps |
| 60 | HP Motors | VFD | 110 amps | 30-50 amps |
| 75 | HP Motors | VFD | 110 amps | 90-150 amps |
| 100 | HP Motors | VFD | 200 amps | CT operated relay |
| 125 | HP Motors | VFD | 200 amps | CT operated relay |
| 150 | HP Motors | VFD | 200 amps | CT operated relay |
| 150 | HP Motors | VFD | 300 amps | CT operated Relay. |
| 200 | HP Motors | VFD | 300 amps | CT operated Relay. |
| 250 | HP Motors | VFD | 400 amps | CT operated Relay. |
| 300 | HP Motors | VFD | 400 amps | CT operated Relay. |
| 400 | HP Motors | VFD | 600 amps | CT operated Relay. |
| 600 | HP Motors | VFD | 900 amps | CT operated Relay. |

Two speed motors when specified, be provided with DOL starter irrespective of it rating.

## 10.0 Cable Trays

Ladder and perforated type Cable Trays be of Hot dip Galvanized type and factory fabricated out of CRCA sheet with standard accessories like tee, bends, couplers etc. for different loads and number and size of cables as given below :

Cable trays be galvanized as per Specifications..

a. 1500 mm wide
Runners 25 x 100 x 25 x 3 mm
Rungs 2# 20 x 40 x 20 x 3 mm 250 mm C/C
Suspenders 2 Nos. 40 x 40 x 5 mm GI angle 1500 mm C/C with base support of 40x 40 x 5mm GI angle.

b. 1200 mm wide
Runners 25 x 100 x 25 x 3 mm
Rungs 2# 20 x 40 x 20 x 3 mm 250 mm C/C
Suspenders 2 Nos. 40 x 40 x 5 mm GI angle 1500 mm C/C with base support of 40x 40 x 5mm GI angle.

c. 1000 mm wide
Runners 25 x 100 x 25 x 3 mm
Rungs 2# 20 x 40 x 20 x 3 mm 250 mm C/C
Suspenders 2 Nos. 40 x 40 x 5 mm GI angle 1500 mm C/C with base support of 40x 40 x 5mm GI angle.

d. 750 mm wide
Runners 20 x 75 x 20 x 2.5 mm
Rungs 20 x 30 x 20 x 2.5 mm 250 mm C/C

Suspenders 2 Nos. 32 x 32 x 5 mm GI angle 1800 mm C/C with base support of 40x 40 x 5mm GI angle.

e 600 mm wide
Runners 20 x 75 x 20 x 2.5 mm
Rungs 20 x 30 x 20 x 2.5 mm 250 mm C/C
Suspenders 2 Nos. 32 x 32 x 5 mm GI angle 1800 mm C/C with base support of 40x 40 x 5mm GI angle.

f. 450 mm wide
Runners 20 x 75 x 20 x 2.5 mm
Rungs 20 x 30 x 20 x 2.5 mm 250 mm C/C
Suspenders 2 Nos. 25 x 25 x 4 mm GI angle 1800 mm C/C with base support of 40x 40 x 5mm GI angle.

g. Supply and fixing of perforated type cable trays of the following sizes of pre-galvanized iron.

i. 600 x 40 x 40 x 2 mm thick
i. 450 x 40 x 40 x 2 mm thick
i. 300 x 40 x 40 x 2 mm thick
ii. 150 x 40 x 40 x 2 mm thick

**Note :** Suitable length of 10 mm dia GI rod suspenders at 1800 mm interval shall be included in the item for perforated type cable tray.

# ANNEXURE - V
## Technical data Sheets

**1.0 VRF UNIT**

**1.1 General**

a. Manufacturer

b. Material and thickness of casing

c. Material and thickness of drain pan.

d. Type of vibration isolator

**1.2 Fan Section**

a. Manufacturer.

b. Type of fan

c. Type of bearings.

d. Fan RPM

**1.3 Motor**

a. Manufacturer

b. Type

c. Motor speed (RPM)

d. Motor Efficiency

e. Class of Insulation

**1.4 Cooling Coil.**

a Manufacturer

b Material of tubes

c Material of fins

d No of fins/inch

**1.5 Air Filters**

a. Manufacturer

b. Type of filters

c. Filter medium

d. Pressure drop across filters (mm. of water) / Clean & Dirty

### 1.6 Operating Data

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AHU NO.

---------------------------------------------------------------------------------------------------------------

AIR QTY (CFM)

---------------------------------------------------------------------------------------------------------------

TOTAL SP (mmWg)

---------------------------------------------------------------------------------------------------------------

FAN ABSORBED POWER (BHP/BkW)

---------------------------------------------------------------------------------------------------------------

FAN SPEED (RPM)

---------------------------------------------------------------------------------------------------------------

FAN OUTLET VELOCITY (FPM or m/s)

---------------------------------------------------------------------------------------------------------------

FAN MOTOR (HP/kW)

---------------------------------------------------------------------------------------------------------------

SOUND POWER LEVEL (DB re 10-12W)

| Mid Freq. | 63Hz | 125Hz | 250Hz | 500Hz | 1000Hz | 2000Hz | 4000Hz | 8000Hz |
|---|---|---|---|---|---|---|---|---|
| SWL(dB) | | | | | | | | |

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SOUND PRESSURE LEVEL(dBA) at fan outlet

| Mid Freq. | 63Hz | 125Hz | 250Hz | 500Hz | 1000Hz | 2000Hz | 4000Hz | 8000Hz |
|---|---|---|---|---|---|---|---|---|
| SPL(dB) | | | | | | | | |

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COIL FACE AREA (Ft2)

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COIL FACE VEL (FPM)

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AIR SIDE PRESSURE DROP ACROSS COOLING COIL (mmWg)

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WATER SIDE PRESSURE DROP IN COOLING COIL (Kg/cm2)

---------------------------------------------------------------------------------------------------------

NUMBER OF ROWS

---------------------------------------------------------------------------------------------------------

WATER VELOCITY IN TUBES (m/s)

---------------------------------------------------------------------------------------------------------

TYPE OF FILTERS

---------------------------------------------------------------------------------------------------------

FILTER FACE VELOCITY (FPM or m/s)

---------------------------------------------------------------------------------------------------------

FILTER EFFICIENCY (%) & PARTICLE SIZE (µm)

---------------------------------------------------------------------------------------------------------

AIR PRESSURE DROP IN CLEAN & DIRTY CONDITIONS (mmWg)

---------------------------------------------------------------------------------------------------------

TYPE OF HEATER (Hot water / electrical)

---------------------------------------------------------------------------------------------------------

HEATING CAPACITY (kW)

---------------------------------------------------------------------------------------------------------

AIR SIDE PRESSURE DROP ACROSS HEATER / HOT WATER COIL (mmWg)

---------------------------------------------------------------------------------------------------------

OVERALL DIMENSIONS
LxWxH (METRES)

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OPERATING WEIGHT